Where Does the Gain Come From? A Controlled Study of Training-Free RoPE Extrapolation to 4K in Diffusion Transformers

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Abstract

Training-free methods let a diffusion transformer trained near one megapixel generate 4096×4096 images by editing its rotary position embedding (RoPE). Methods are often compared as complete packages, leaving the contribution of each edit unclear. On FLUX.1-Krea-dev, over 187 held-out prompts with paired bootstrap intervals, SEGA's base and global temperature alone recover all of its ImageReward gain over an NTK base (104%, interval [97, 113]). They tie the released method on all three reward models and crop MUSIQ, where "tie" means that the difference interval covers zero, not equivalence. The per-band allocation adds 2.09 full-frame MUSIQ points and reduces excess fine-scale power. Four per-band laws that we derive from efficient coding and contrast constancy do no better: the two evaluated held out tie or lose to SEGA on every reward and image-quality metric, and trajectory diagnostics suggest why. A development comparison also finds a halftone lattice on FLUX.1-dev under official DyPE code but not on FLUX.1-Krea-dev; differing guidance prevents attributing this contrast to the checkpoint alone. SEGA's printed formulation and released code differ at each of the six stages we compared. Finally, 224-pixel reward models cannot directly resolve the top 4 octaves of a 4K image, and the default flow-shift extrapolation collapses the noise schedule. Claims for spectral or content-adaptive RoPE edits should be tested against a base-plus-temperature control with full-resolution metrics.

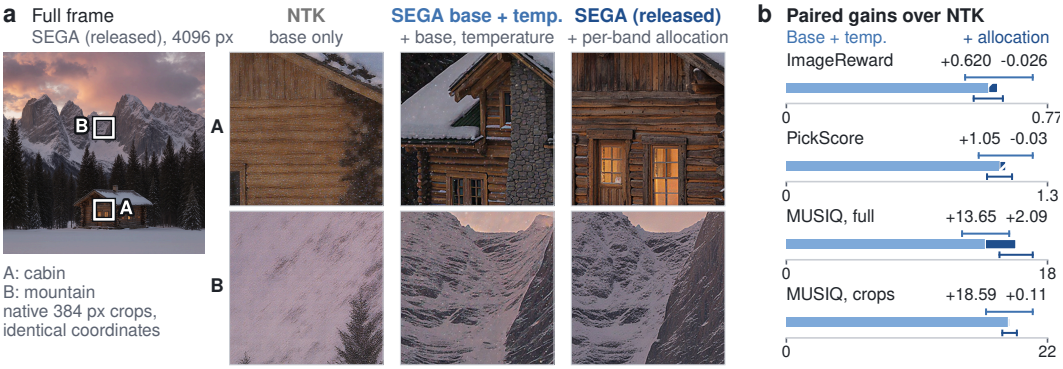


Figure 1: SEGA's 4K gain over the NTK base comes mostly from its RoPE base and global temperature. (a) Held-out row 32, seed 0, chosen by its cabin subject during revision before viewing that row's outputs. The illustrative example is not random. After viewing, we fixed shared native 384^2 crops centred at $(2048, 3200)$ and $(2048, 1536)$ to show the cabin and mountain across arms. (b) Signed absolute gains from base and temperature (SEGA base + temp. minus NTK) and from the allocation (SEGA (released) minus SEGA base + temp.), with a separate axis per metric and 95% paired bootstrap intervals over 187 held-out prompts. Allocation intervals are translated by the base-and-temperature point estimate for stacking; they are not intervals for the total gain. Hatching marks negative allocation estimates.

1 Introduction

Diffusion transformers trained at about one megapixel are routinely asked for 4096×4096 images, 16 times more tokens than any training image. Training-free methods make this work by editing the rotary position embedding (RoPE) [47] at inference: they rescale its frequencies [7, 10, 41], schedule the rescaling with the noise level [27], sharpen attention with a global logit temperature [28, 37] and, most recently, scale each rotary band by a gain read from the spectrum of the current latent [45]. Each method is compared as a package with the previous one, and the improvement is credited to its newest component.

We ask where the improvement comes from. We take SEGA [45], a strong recent method with released code, on the checkpoint its authors use (FLUX.1-Krea-dev), and compare its base and global temperature together against its content-adaptive per-band allocation of query and key gains. All arms share prompts, seeds, schedule and sampler; the dev/held-out split was fixed before any held-out score was seen; every main comparison is paired over 187 held-out prompts with bootstrap intervals, and a seed-1 replication covers 55 of them. We then test whether per-band laws with a principled basis beat SEGA's heuristic: from efficient coding and contrast constancy we derive four controllers, use the dev set to choose variants, and evaluate the survivors held out. A written advancement gate applied from the second campaign onward. Our findings, in order of strength:

(i) **Base and temperature carry most of the gain** (Figure 1). Of SEGA's gain over the NTK base, its base and global temperature alone carry 104% on ImageReward, 103% on PickScore and 99% on crop MUSIQ. The per-band allocation adds a full-frame MUSIQ gain (+2.09 [+0.96, +3.25]) alongside less excess fine-scale power, with no detected reward-model gain (Section 5.1). MUSIQ is a learned quality proxy, not a sharpness measurement.

(ii) **Principled per-band laws do not beat the heuristic.** The two controllers that reached the full held-out set tie or lose to SEGA (released) on every reward and image-quality metric; the closer one, SEGA + TM, ties it on ImageReward and full-frame MUSIQ and loses on PickScore (-0.08) and crop MUSIQ (-1.13). Trajectory diagnostics suggest possible causes: a gate that opens late, a target that improves spectral match while lowering MUSIQ, an allocation that is mostly a fixed tilt and a prompt-dependent response to per-band gain changes (Section 5.2).

(iii) **Checkpoint comparisons need matched settings.** In the development comparison, FLUX.1-dev renders a halftone lattice at 4096^2 even under official DyPE code, while FLUX.1-Krea-dev does not. Guidance also differs, so the contrast does not isolate a checkpoint effect (Section 5.3).

(iv) **SEGA's paper and code differ.** The released code changes the sensor input, the gain law, its bounds and the frequency base. In a descriptive eight-prompt pilot the printed equations score worse than the released code on every metric, and their static tilt destroys images (Section 5.4).

(v) **4K evaluation has traps.** 224-pixel reward models cannot see the top 4 octaves of a 4K image; the default flow shift extrapolates to $\mu = 11.55$, so every model call sees almost pure noise; and forced FlashAttention-2 takes 1.54 times as long as cuDNN attention on a GH200 (Section 5.5).

We contribute the controlled decomposition, negative results with diagnostic evidence, and a protocol that makes these claims checkable. Code, per-image scores and the scripts that generate every table and number in the paper accompany the paper.

2 Background

RoPE in FLUX. FLUX [5] packs the autoencoder latent into 2×2 patches, so one token covers 16×16 pixels and a 4096^2 image has a 256^2 token grid (64^2 at 1024^2). Each 128-dimensional attention head has rotary blocks of 16, 56 and 56 dimensions for an index axis and the two spatial axes. Pair j of spatial axis a rotates at $\omega_{a,j} = \theta^{-2j/D}$ with $\theta = 10^4$ and $D = 56$, so wavelengths $2\pi/\omega_{a,j}$ run from about 6 to about 45,000 tokens over the 28 pairs.

The logit as a Fourier series over the offset. Write the query and key coefficients of pair j on axis a as complex numbers $\tilde{q}_{a,j}$ and $\tilde{k}_{a,j}$, and let a method multiply both by a gain $m_{a,j}$ after the rotation,

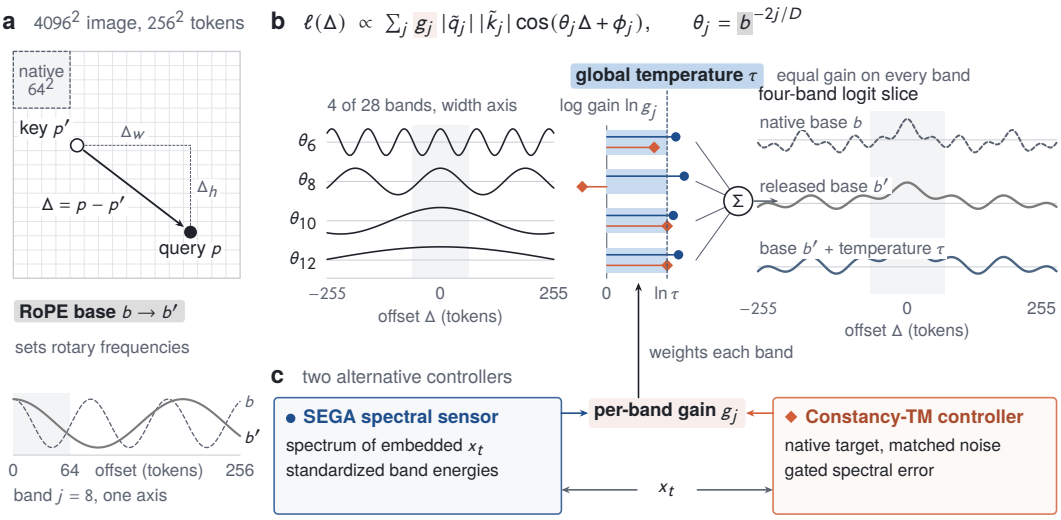


Figure 2: The RoPE base changes frequencies, a global temperature scales all terms, and per-band gains reweight individual terms. (a) Query, key and offset on the token grid of a 4096^2 image; the inset shows band 8 under the native base b (dashed) and SEGA's released base b'. (b) Four of the 28 width-axis bands with their total log gains (sky: $\ln\tau$; circles: SEGA (released); diamonds: Constancy-TM) from one held-out sample at step 14. The three separate traces sum these four bands with unit content amplitudes and zero phases on the same vertical scale; they illustrate the base and temperature changes, not measured query–key logits. (c) Two alternative controllers set the total gain g_j, which already includes temperature, from their respective spectral measurements.

with which it commutes. The logit between a query at position u and a key at position v is then

$$\ell_{uv} = \frac{1}{\sqrt{d_k}} \sum_{a,j} g_{a,j} \left|\tilde{q}_{a,j}\right|\left|\tilde{k}_{a,j}\right| \cos\!\left(\omega'_{a,j}\Delta^a_{uv} + \phi_{a,j}\right), \qquad g_{a,j} = m^2_{a,j}, \tag{1}$$

a Fourier series in the offset $\Delta^a_{uv} = p^a_u - p^a_v$ with one term per rotary pair, plus a non-rotating index-axis part. Content sets the magnitudes and phases $\phi_{a,j}$; extrapolation changes only the frequencies $\omega'_{a,j}$ and the gains $g_{a,j}$. A gain shared by all pairs, $g \equiv \tau$, is an inverse temperature; a j-dependent gain reweights the bands of the kernel.

Three knobs. We sort methods by the knobs they turn (Figure 2). *Frequency base*: at 4096^2 the offsets reach $s = 4$ times the native range, so slow pairs see phases never seen in training. Position interpolation divides every frequency by s [10]; NTK-aware scaling raises the base to $\theta' = \theta s^{D/(D-2)}$, which leaves the fastest pair alone and divides the slowest by s [7]; YaRN interpolates only the slow pairs [41]. *Global temperature*: with N keys instead of N_{train}, softmax entropy grows roughly with $\log N$, and multiplying logits by $\tau_N = \log N / \log N_{\text{train}}$ offsets it [28]; at 4096^2, $\tau_N = 1.333$. *Per-band gain*: a j-dependent $g_{a,j}$, possibly varying with the noise level and with the image.

The flow schedule at 4K. FLUX samples a rectified flow [34, 35] with a time shift $\sigma' = e^\mu / \left(e^\mu + 1/\sigma - 1\right)$ whose μ grows linearly with the token count [16]. The diffusers default [49] extrapolates that line to $\mu = 11.55$ at 4096^2. The last evaluated noise level is then $\sigma = 0.99974$: every model call sees almost pure noise, and the final Euler step jumps from there to the image. The official DyPE and SEGA pipelines cap μ at 1.15 (last $\sigma = 0.10472$), as do all our comparison arms. This is a numerical schedule diagnosis; we did not measure its effect on image quality. Comparisons that mix the two schedules also confound position embedding with sampling.

3 Methods under study

All arms share the sampler and differ only in ω' and g of Equation (1) (Table 1; exact formulas and code revisions in Appendix E).

Table 1: Arms and the knobs they turn. Temperature is the logit gain at 4096^2; t/x: varies with the noise level / with the image. Held-out arms ran on rows 8–194, the pilot and dev arms only on the eight dev prompts.

Arm	Frequency base	Temperature	Per-band gain	t/x	n
NTK	$\theta\, s^{2D/(D-2)}$	none	none	no / no	187
DyPE	PI/NTK ramps in σ	1.30 (spatial)	none	yes / no	187
SEGA (released)	$\theta\, s^{2D/(D-2)}/(1{+}0.1 \ln s)$	$s^{0.16} = 1.248$	sensor, ≥ 1	yes / yes	187
SEGA (paper eqs.)	$\theta\, s^{2D/(D-2)}$	$s^{0.16}$	sensor, unbounded	yes / yes	8 (pilot)
SEGA base + temp.	as SEGA (released)	$s^{0.16}$ on all dims	none	no / no	187
Constancy	as NTK	$\tau_N = 1.333$	gated, late target	yes / yes	41 (partial)
Constancy-TM	as SEGA (released)	$s^{0.16}$	gated, time-matched	yes / yes	187
Alias-T	as SEGA (released)	none	alias count, static	no / no	8 (dev)
SEGA + TM	as SEGA (released)	$s^{0.16}$	SEGA × time-matched	yes / yes	187

Table 2: Our reading of SEGA's printed formulation differs from the release at each of the six stages compared. Released revision `ddaab08552c7`; source locations, equation numbers and interpretation limits are in Appendix E. Here s is the scale per side and $D = 56$.

Stage	Printed equations	Released code
Sensor input	channel mean of the latent	channel mean of the projected input features
Pair-to-bin map	physical frequency, clipped	rank of log period
Allocation	$m_{\text{ref}}(1 - \varsigma u)$, $u = \tanh z - \overline{\tanh z}$	$\max\big(1,\, m_{\text{ref}}(1 - 0.15\,\varsigma c)\big)$, $c = \tanh 1.5z - \overline{\tanh 1.5z}$
Bounds	none	gains clamped at 1
Index axis	m_{ref}	1
Frequency base	$\theta s^{2D/(D-2)}$ per axis	$\theta s^{2D/(D-2)}/(1 + 0.1 \ln s)$, longest side

Frequency baselines. NTK is the strengthened NTK base that SEGA builds on, $\theta' = \theta s^{2D/(D-2)}$, which divides the slowest pair's frequency by s^2. With no temperature and no allocation, it is the floor of the decomposition. DyPE [27], ported from the official code, blends position interpolation, NTK and native frequencies with two ramps in σ^2, so fine pairs return to their native frequencies late in denoising; its cos/sin amplitude $1 + 0.1 \ln s$ acts as a logit gain of 1.30 on the spatial dimensions.

SEGA: paper equations and released code. SEGA [45] reads the spectrum of its input at every model call and sets per-pair gains around a reference gain $m_{\text{ref}} = s^{0.08}$. In the printed equations, the energy E_a of the channel-mean latent at pair j's frequency bin is standardized across pairs, $z_{a,j}$, and

$$m_{a,j} = m_{\text{ref}}\big(1 - \varsigma\, u_{a,j}\big), \qquad u_{a,j} = \tanh z_{a,j} - \text{mean}_i \tanh z_{a,i}, \qquad \varsigma = 1 - \text{SF}(E_{\text{iso}})^{1.5}, \tag{2}$$

where SF is the flatness of the radial spectrum. Low-energy pairs, usually the fast ones, thus gain attention weight in proportion to how far the spectrum is from flat. Our reading of the printed formulation differs from the release at each of the six stages in Table 2. We call its parity-tested port SEGA (released) and our implementation of the printed equations SEGA (paper eqs.); static variants use the mean allocation over pilot prompts and steps.

SEGA base + temp.: the decomposition control. SEGA base + temp. keeps the frequency base of SEGA (released) and applies its reference logit gain $s^{0.16}$ uniformly to every dimension. It differs from SEGA (released) in three ways: no per-band shape, no lift in mean gain from the clamp at 1, and a gain of $s^{0.16}$ rather than 1 on the index axis.

3.1 Per-band laws from vision science

SEGA's gain law is a heuristic; we asked whether laws from biological vision do better. Perceived contrast in each spatial-frequency band stays nearly constant above threshold across viewing distances, and so across sampling rates [20]. Efficient coding explains this by weighting each band by its visibility against noise, the Wiener factor $S/(S+N)$ [1, 2, 48], and by renormalizing band contrast above threshold toward natural-image statistics [18, 39, 51]. A diffusion sampler knows what the retina does not: the exact noise power in every band at every step. Each law sets $\ln g_{a,j} = \ln \tau + \Delta_{a,j}$, a temperature plus a bounded per-band correction.

Constancy. For half-octave bands b, a gate compares the current input's band energy with the known noise floor, $\rho_{a,b} = \max\big(E_a^x[b]/(\sigma^2 C) - 1, 0\big)$ and $w_{a,b} = \rho_{a,b}/(1 + \rho_{a,b})$, and an error compares

the previous clean estimate's band energy $\hat{E}_a[b]$ with a native-resolution target P^{nat}, the mean log spectrum of the model's own 1024^2 samples on 16 calibration prompts outside the evaluation set:

$$e_{a,b} = \tfrac{1}{2}\big(P^{\text{nat}}_a(f_b) + c_a - \ln \hat{E}_a[b]\big), \qquad \delta_{a,b} = \eta\, w_{a,b}\, \text{clip}(e_{a,b}, -1, 1), \tag{3}$$

where c_a anchors the target level on visible bands of 16 to 64 token wavelengths. The pair correction $\Delta_{a,j}$ interpolates δ in $\log f$, with $\tau = \tau_N$ and $\eta = 1$. A natural spectrum gives $e = 0$ and reduces the law to the global temperature. The target is the native spectrum of the final, nearly clean steps.

Constancy-TM. The same law with a time-matched target: $P^{\text{nat}}(f, \sigma)$ is the clean estimate's native spectrum at the same noise level, tabulated over all 28 native steps, so the anchor no longer needs the gate. Built on the base and temperature of SEGA (released), it isolates the allocation, with SEGA base + temp. as its control.

Alias-T and SEGA + TM. Alias-T applies the log-N temperature pair by pair, taking as a pair's key pool the number of keys it can resolve on the grid (Appendix E); Alias-T-s shifts all its log gains by one constant to match SEGA's reference gain in geometric mean. SEGA + TM multiplies the allocation of SEGA (released) by the time-matched correction, $\ln g = \ln g^{\text{SEGA}} + \Delta$, with $\eta \in \{0.5, 1\}$ on the dev set and $\eta = 1$ held out.

4 Experimental protocol

Model and sampler. The held-out comparisons use the FLUX.1-Krea-dev transformer [6], the checkpoint loaded by the official DyPE and SEGA scripts, with the FLUX.1-dev text encoders and autoencoder [5] (Appendix E). A 4096^2 image has 65,536 tokens, 16 times the 4,096 of a native 1024^2 image ($s = 4$ per side). Every held-out arm uses 28 Euler flow-matching steps, distilled guidance 4.5, bfloat16, batch size one, the same per-prompt seeds and the capped flow shift $\mu = 1.15$ (Section 2).

Prompts and the dev/held-out split. Of the 195 evaluation captions of Aesthetic-4K [55], rows 0–7 form a development set for variant and hyperparameter choices, and rows 8–194 the held-out set ($n = 187$) for the main comparisons. We fixed this split after an eight-prompt pilot and before inspecting any held-out score. The decomposition hypothesis arose from the dev pilot and was tested only held out. From the second campaign on, a written gate decided which variants could leave the dev set: against SEGA (released) on the eight dev prompts, a variant needed higher ImageReward and full-frame MUSIQ and a CLIP drop of at most 0.3, and at most two variants could advance per campaign. Dev numbers are selection evidence only (Appendix D.2). Seed 0 is used throughout; a seed-1 replication covers held-out rows 8–62 ($n = 55$) for SEGA (released), SEGA base + temp. and Constancy-TM. This replication and the partial Constancy run use the first available rows of the fixed manifest, not random prompt subsamples.

Metrics. *Alignment and preference*: CLIP score [24, 43], the scaled cosine $100 \max(\cos, 0)$ of ViT-L/14 embeddings, PickScore [32] and ImageReward [53]. These models resize their input to 224 pixels, so we read them as measures of layout and alignment (Section 5.5). *No-reference quality*: MUSIQ [30] and CLIP-IQA [50], on the full 4096^2 frame ("full-frame") and averaged over four fixed 1024^2 crops shared by all methods ("crop"). The pyiqa 0.1.16 MUSIQ pipeline combines 224- and 384-pixel views with every nonoverlapping 32^2 native-resolution patch: 16,577 patches for a 4096^2 frame, with no native-patch cap (Appendix E). Thus full-frame MUSIQ includes all original pixels but also differs from crop MUSIQ in spatial context and scale weighting. *Spectral fidelity and repetition*: the root-mean-square and log-log slope errors of luminance power in 11 octave bands from 2^{-12} to 2^{-1} cycles per pixel, against the mean profile of the 195 Aesthetic-4K reference images, and a tiling index, the autocorrelation at a 1024-pixel lag minus that at 1536 pixels. FID and KID [4, 25] are too noisy at our sample sizes to rank methods (Appendix B). No metric here measures 4K quality on its own, and we ran no human study.

Statistics. Each comparison is a paired difference against SEGA (released) over the same prompts and seed, with a 95% percentile bootstrap interval over 2,000 prompt resamples. Better, worse and tie describe these nominal intervals: a tie means the interval covers zero, not equivalence. We also

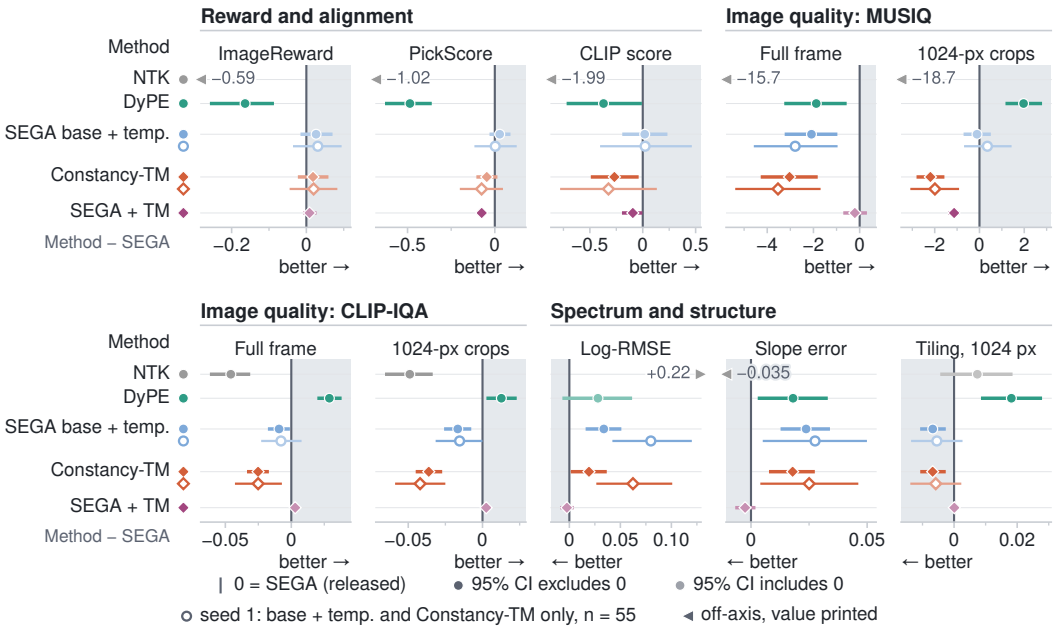


Figure 3: No per-band law improves on SEGA (released) in any reward or image-quality metric. Paired held-out differences from SEGA (released) with 95% bootstrap intervals (rows 8–194, seed 0, $n = 187$). Hollow markers: seed-1 replication on rows 8–62 ($n = 55$) for SEGA base + temp. and Constancy-TM; SEGA + TM has no seed-1 replication. Shaded half: better direction. Isolated gains, such as crop MUSIQ and CLIP-IQA for DyPE, come with losses elsewhere.

compute approximate two-sided centered-bootstrap p-values from the absolute null-centered mean differences and apply Holm correction separately across the paired comparisons in Table B.1 and in Table B.4. Stars mark adjusted $p \leq 0.05$; intervals and colour remain nominal. The allocation's full-frame MUSIQ gain survives both families, whereas its CLIP-IQA gain and SEGA + TM's small CLIP loss do not. These bootstrap tests provide approximate, rather than exact finite-sample, error control. The remaining descriptive tables retain nominal intervals.

5 Results

5.1 Decomposing SEGA (released): base and temperature do most of the work

Figure 3 compares every held-out arm with SEGA (released); Table B.1 in Appendix B gives the numerical intervals. Two rows carry the decomposition. NTK keeps SEGA's family of frequency base and drops both the temperature and the allocation; it is far worse on every reward and image-quality metric (ImageReward −0.594, full-frame MUSIQ −15.74, crop MUSIQ −18.70). SEGA base + temp. keeps SEGA's released base and global temperature and drops the allocation, with the index-axis difference specified in Section 3. It ties SEGA (released) on ImageReward (+0.026 [−0.016, +0.070]), PickScore, CLIP and crop MUSIQ, and falls behind on full-frame MUSIQ (−2.09 [−3.25, −0.96]), CLIP-IQA and the spectral error.

Figure 1b shows the paired absolute gains, and Table B.2 (Appendix B) expresses them as shares. Of SEGA's gain over NTK, base and temperature carry 104% of the ImageReward gain (interval [97, 113]), 103% of PickScore, 101% of CLIP and 99% of crop MUSIQ. The allocation's own contribution, SEGA (released) minus SEGA base + temp., is +2.09 [+0.96, +3.25] on full-frame MUSIQ and +0.009 [+0.001, +0.018] on CLIP-IQA; its intervals cover zero on all three reward models and on crop MUSIQ. Its higher full-frame MUSIQ accompanies lower excess luminance power at 8 to 32 pixel wavelengths (Figure 4). This spectral change is consistent with suppressing fine-scale artefacts, but it does not establish what caused MUSIQ to rise. Neither spectral power nor MUSIQ alone measures perceived sharpness.

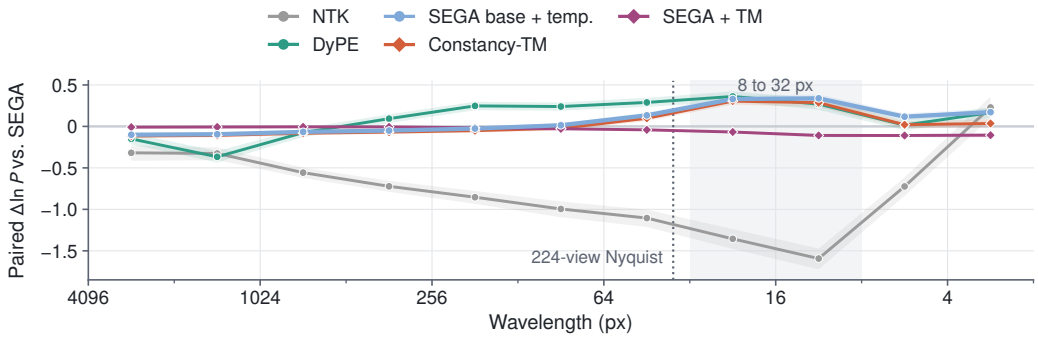


Figure 4: Removing the allocation leaves more fine-scale power than released SEGA. Paired differences in octave-band log luminance power from SEGA (released), over held-out rows 8–194 ($n = 187$, seed 0), with 95% prompt-bootstrap intervals. The excess at 8 to 32 pixel wavelengths lies above the marked Nyquist frequency of a 224-pixel view. This spectral contrast accompanies the full-frame MUSIQ gap; it does not isolate its cause. Figure A.3 also shows errors against the reference profile.

DyPE makes a different trade. Against SEGA (released) it is worse on all three reward models (ImageReward −0.164 [−0.258, −0.087], PickScore −0.49) and better on crop MUSIQ (+1.97 [+1.16, +2.80]) and CLIP-IQA (+0.029). No single metric orders the methods, so we report all of them.

Replication at seed 1. On the 55 seed-1 prompts, SEGA base + temp. again ties SEGA (released) on ImageReward, PickScore and CLIP and loses on full-frame MUSIQ (−2.79 [−4.58, −0.95]). 17 of the 20 tie/better/worse verdicts for SEGA base + temp. and Constancy-TM agree with seed 0 on the same prompts (Table C.1).

Qualitative. Figure 5 shows the first two held-out prompts, with shared crop coordinates chosen to retain detail in every arm. The compositions still differ, so these crops illustrate visible differences rather than a controlled perceptual comparison. Figure A.1 (Appendix A) retains the original three-prompt grid.

5.2 Principled per-band laws do not beat SEGA (released)

Outcome. Neither law evaluated on the full held-out set improves on SEGA (released) in any reward or image-quality metric (Table B.1, bottom block). Constancy-TM ties it on ImageReward and PickScore and is worse on CLIP (−0.27), full-frame MUSIQ (−3.04), crop MUSIQ (−2.19), CLIP-IQA and the spectral error; its one gain is a lower tiling index (−0.0068). It also loses to SEGA base + temp., which shares its base and temperature and has no allocation: crop MUSIQ −2.08 [−2.43, −1.75] and full-frame MUSIQ −0.95, against small gains in both spectral errors (−0.015 and −0.006). SEGA + TM at $\eta = 1$, the only variant to pass the campaign-2 dev gate, ties SEGA (released) on ImageReward (+0.009 [−0.008, +0.027]) and full-frame MUSIQ (−0.21 [−0.71, +0.32]) and loses on PickScore (−0.08 [−0.10, −0.05]) and crop MUSIQ (−1.13 [−1.39, −0.90]); its CLIP interval, [−0.20, −0.00], ends just below zero. Alias-T and Alias-T-s failed the gate on ImageReward and were never run held out (Table D.1); the original Constancy has only a partial held-out run (Appendix D). Four diagnostics, on dev prompts and on the logged gains of held-out runs, suggest possible explanations (Figure F.1 in Appendix F). They do not identify causal mediators of image quality.

The visibility gate opens too late. Constancy's correction is exactly zero until a band with a wavelength of 16 to 64 tokens holds more than twice the noise power ($w > \frac{1}{2}$). On the eight dev prompts it first becomes nonzero at $\sigma = 0.60$ and is nonzero on all of them only by $\sigma = 0.46$ (Figure F.1a). For most of sampling, including the steps that fix the layout, Constancy is thus the NTK base with a log-N temperature: the gate is principled, but at 4096^2 the bands it guards stay below the noise floor until late.

A native-resolution target improves spectral match while lowering MUSIQ. The time-matched target opens the gate early. On the held-out Constancy-TM runs, corrections exceed 0.1 in log gain

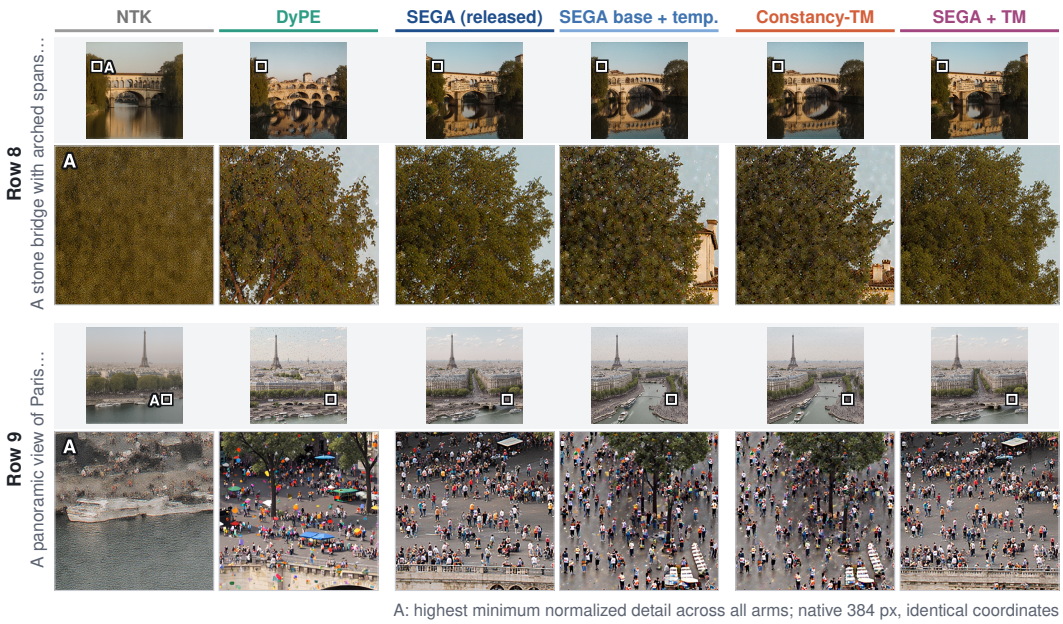


Figure 5: Native-resolution crops expose differences hidden by a small overview. The first two held-out prompts (rows 8 and 9), seed 0, with a full-frame thumbnail and an unaltered 384-pixel crop for each arm. During review, we selected one shared crop location per row by maximizing the minimum normalized Laplacian energy across all six arms on a fixed grid. Each arm's energies are normalized by its 90th percentile over candidate locations. This output-dependent rule favors locations containing detail in every arm; it does not select a winning method. Prompts were unchanged, and differing compositions remain a limitation.

after the first 8 steps (median), and 70% of all nonzero corrections are negative: they lower the mid pairs 6 to 8 (-0.24 at the last step) and, late, the finest pairs 0 to 2 (-0.22). These corrections reduce the gains of the corresponding rotary bands. Against SEGA base + temp., output spectral errors shrink slightly while both MUSIQ scores fall. Matching a native spectrum and maximizing a learned quality score are different objectives; the measurements do not show which individual gain changes cause the score losses.

SEGA's allocation is mostly a fixed tilt. Across the held-out prompts the released logit gain stays between 1.00 and 1.61. A fixed tilt over pairs explains 56% of its variance and a tilt plus a per-step profile 80%. The prompt-dependent remainder, 20% overall and 7.3% from step 14 on, is an upper bound on content dependence, because it also contains each prompt's noise draw. At the last step the finest pair has a logit gain of 1.47 and the 6 slowest pairs per axis sit at the clamp of 1. A law that adapts more to content therefore has little to copy: SEGA + TM raises the prompt-dependent share to 37% and gains nothing held out. A CPU study of the printed equations finds the same structure on another latent space (Appendix F).

The response to gain changes is weak and prompt-dependent. Every law assumes that raising a band's gain raises its output power. A probe on three dev prompts that scales one octave's gain by 1.5 (Figure F.1b) finds a clear same-octave width-axis response for one prompt and a near-zero one for the other two, off-diagonal responses spread across octaves, and a weak height-axis response. On two dev prompts, a local linear model predicts half-step band-power changes little better than no change (Appendix F). These probes question a simple band-local response model, but their small sample does not establish a universal limit on per-band control.

5.3 Checkpoint comparisons need matched settings

Our first campaign used FLUX.1-dev, on which DyPE and Constancy produced a dense halftone lattice over the 4096^2 frame and NTK a featureless blur. An official-code DyPE check reproduced the lattice on dev prompt 0, seed 42, at guidance 3.5; the corresponding FLUX.1-Krea-dev run at guidance 4.5 had no visible lattice (Figure 6). This rules out our port as a necessary cause of that par-

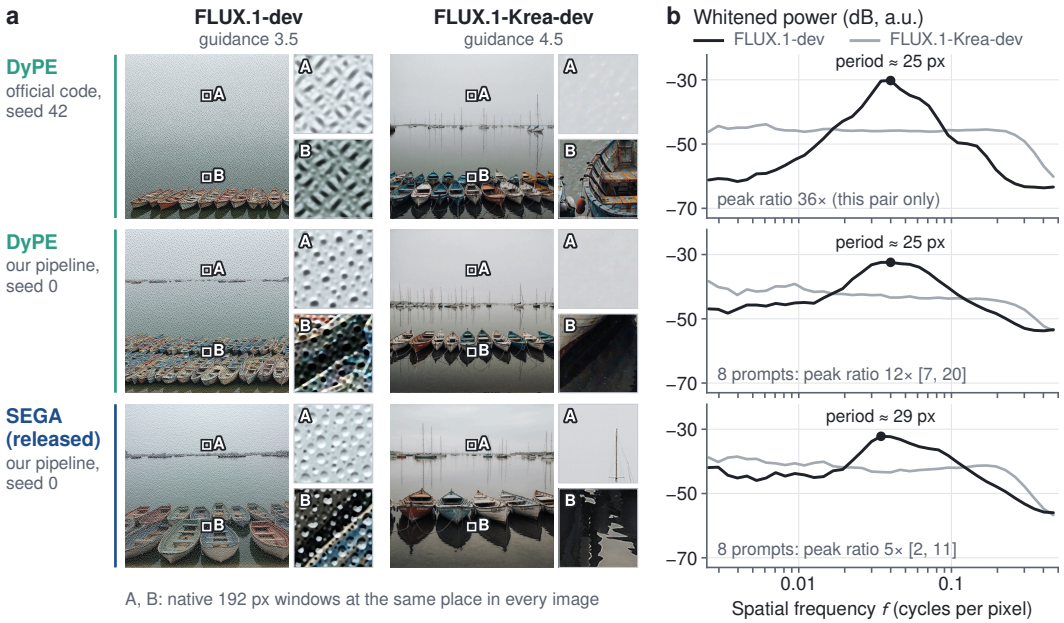


Figure 6: The development comparison shows a lattice on FLUX.1-dev, but checkpoint and guidance both change. (a) Dev prompt 0 under official DyPE code (seed 42) and our DyPE and SEGA (released) pipelines (seed 0), at guidance 3.5 (FLUX.1-dev) and 4.5 (FLUX.1-Krea-dev); A and B are native 192-pixel windows at the same place in every image. (b) Whitened radial power spectra of luminance: FLUX.1-dev peaks at the lattice period and FLUX.1-Krea-dev does not in these samples. Official-code evidence covers one prompt per checkpoint; the descriptive eight-prompt peak summaries use our pipeline. The figure does not isolate a checkpoint effect.

ticular failure, but the guidance difference prevents isolating the checkpoint. DyPE's main reported experiments use FLUX.1-Krea-dev [27], as do the official scripts we inspected. We used FLUX.1-Krea-dev for all held-out comparisons and retain the FLUX.1-dev pilot as descriptive evidence only.

5.4 SEGA: paper versus code

With the printed law (Equation (2) and Table 2), the allocation is unbounded, and at 4096^2 the physical bin map sends every pair with a wavelength above 256 tokens, 19 of the 28 per axis, to the same frequency bin. The released law scales the spread by 0.15 and clamps the result at 1, and its logit gain stays between 1.00 and 1.61 on our held-out runs. In the eight-prompt dev pilot (descriptive only; Appendix D.3), our implementation of the printed equations scores worse than the released code on every metric; freezing the printed allocation into a static tilt destroys the images, while freezing the released allocation costs much less (Figure A.2). Readers who reimplement SEGA from the paper will not reproduce the released method, and comparisons against SEGA should state which version they ran.

5.5 Pitfalls in 4K evaluation

Two further details limit what 4K numbers can show, and what they cost.

224-pixel scorers miss the finest octaves. CLIP, PickScore and ImageReward see a 224-pixel view, whose Nyquist frequency is 0.0273 cycles per original pixel ($2^{-5.2}$); bands 7 to 10 of our 11 octave bands lie entirely above it. These are bands where the arms differ (Figure 4 and Table B.5): in band 7, NTK falls far below the reference profile (log-power error −1.30), the two arms with SEGA's allocation stay closest to it (SEGA (released) +0.06, SEGA + TM −0.01), and SEGA base + temp. carries excess power (+0.39). The limited view offers one possible explanation for the reward-model ties alongside a full-frame MUSIQ gap. Resizing can retain indirect effects or alias fine patterns, but these scorers cannot directly resolve the original fine detail. They are insufficient alone for a 4K quality comparison.

Attention kernel. On the same GH200 node and method, forcing FlashAttention-2 makes a 4096^2 sample take 1.54 times as long as cuDNN-first SDPA (medians 290 s and 188 s), while method families on one node differ by at most 0.6%; one GB200 GPU is 2.01 times faster than a GH200 (Table G.1). Timings of different methods are comparable only with the kernel and the GPU fixed. Our records did not log the dispatched kernel or a hash of the loaded weights; we recovered both from code revisions and task scripts (Appendix G).

6 Discussion and limitations

What to report. A method that changes the rotary base, the temperature and per-band gains at once should report a control that keeps its base and temperature and drops the rest; here that control leaves a gap of 2.09 full-frame MUSIQ points and no detected reward-model gain. The allocation reduces excess fine-scale power, but the association with higher MUSIQ does not establish a perceptual mechanism. Claims that credit spectral or content adaptivity should be tested against this control with full-resolution metrics, because 224-pixel scorers cannot directly resolve the affected detail.

Small dev sets mislead. Eight dev prompts were enough to reject clearly worse variants and too few for anything else. On the dev set SEGA base + temp. looked slightly better than SEGA (released) on MUSIQ, yet held out it trails by 2.09 points; the one variant that passed the second gate lost on PickScore and crop MUSIQ held out; and a rerun of SEGA (released) on the same dev prompts moved MUSIQ by more than the margin of the one variant that passed the third gate (Appendix D.2). The held-out set has now been inspected and cannot serve as fresh confirmation for new variants.

Limitations. We study one model family (FLUX.1-Krea-dev) at one square size, 4096^2; other aspect ratios, sizes, backbones and video are untested. The development comparison in Section 5.3 changes both checkpoint and guidance. All metrics are automatic: the reward models see 224 pixels, MUSIQ and CLIP-IQA are learned proxies with their own biases, the spectral error uses a mean reference profile rather than a per-prompt target, and we ran no human study. MUSIQ includes the full native patch grid, but its full-frame and crop scores also differ in context and scale weighting. Most arms have one seed; the seed-1 replication covers two arms on the first 55 held-out prompts. The main paired tables include approximate Holm-adjusted tests, but intervals remain nominal and the descriptive tables have no multiplicity correction; Table B.4 alone holds 60. The original Constancy has only a partial held-out run. The paper-versus-code numbers come from an eight-prompt dev pilot and are descriptive, the mechanism probes use three dev prompts or fewer, and the CPU study of SEGA uses another latent space. We port SEGA and DyPE from their released code with parity tests and ran the official DyPE code only for Section 5.3. A 4096^2 sample takes a median of 188 s on a GH200 (Appendix G), which limited the number of arms, seeds and resolutions.

7 Related work

Training-free high-resolution generation. U-Net methods dilate convolutions in space [22] or frequency [26], fuse patches [14] or scales [42], restructure features [56] or cascade over frequencies [57]. For rectified-flow transformers [5, 16, 40], I-Max [15] and HiFlow [8] guide the high-resolution trajectory with a low-resolution one, HRDiT [54] adapts pretrained transformers, and Diffusion-4K [55] fine-tunes at 4K and supplies our prompts. We study the single-pass, embedding-level family, which changes no weights and adds no passes.

Rotary extrapolation and attention temperature. Language models remap rotary frequencies by interpolation [10], NTK-aware scaling [7], YaRN [41] or LongRoPE [12]; Barbero et al. [3] and DoPE [52] analyse which frequencies carry position and which carry semantics. For diffusion transformers, RIFLEx [58] lowers the frequency that causes repetition in long videos, DyPE [27] schedules the remapping with the noise level, UltraImage [59] revisits frequency selection, TIDE [36] adds step-aware temperature control, SEGA [45] adds per-band gains, and AReA [44], from SEGA's authors, adapts attention locality. Scaling logits by the log of the key count keeps attention entropy roughly constant, as in variable-size diffusion [28] and Scalable-Softmax [37]; our results suggest that this knob, together with the base, does most of the work at 4096^2.

Spectral views and efficient coding. Diffusion denoises roughly coarse to fine [11, 17], and denoisers adapt to image structure through harmonic bases [21, 29]. Spectral guidance steers toward target spectra [31], CSFlow weights flow matching by contrast sensitivity [19], and Attention Frequency Modulation edits the Fourier bands of cross-attention [38]. Our controllers act on the rotary bands of self-attention and draw on natural-image spectra [13, 18], human sensitivity [33, 39], efficient coding [1, 2, 48], contrast constancy [20] and blur adaptation by divisive gain control [9, 23, 46, 51].

References

[1] Joseph J. Atick and A. Norman Redlich. Towards a theory of early visual processing. *Neural Computation*, 2(3):308–320, 1990.

[2] Joseph J. Atick and A. Norman Redlich. What does the retina know about natural scenes? *Neural Computation*, 4(2):196–210, 1992.

[3] Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! What makes rotary positional encodings useful? In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.06205.

[4] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations (ICLR)*, 2018.

[5] Black Forest Labs. FLUX.1. https://github.com/black-forest-labs/flux, 2024.

[6] Black Forest Labs and Krea AI. FLUX.1 Krea [dev]. https://huggingface.co/black-forest-labs/FLUX.1-Krea-dev, 2025.

[7] bloc97. NTK-Aware scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation. Reddit, r/LocalLLaMA, 2023.

[8] Jiazi Bu et al. HiFlow: Training-free high-resolution image generation with flow-aligned guidance, 2025. arXiv:2504.06232.

[9] Matteo Carandini and David J. Heeger. Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1):51–62, 2012.

[10] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation, 2023. arXiv:2306.15595.

[11] Sander Dieleman. Diffusion is spectral autoregression. Blog post, https://sander.ai/2024/09/02/spectral-autoregression.html, 2024.

[12] Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.13753.

[13] Dawei W. Dong and Joseph J. Atick. Statistics of natural time-varying images. *Network: Computation in Neural Systems*, 6(3):345–358, 1995.

[14] Ruoyi Du, Dongliang Chang, Timothy Hospedales, Yi-Zhe Song, and Zhanyu Ma. DemoFusion: Democratising high-resolution image generation with no $$$. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2311.16973.

[15] Ruoyi Du, Dongyang Liu, Le Zhuo, Qin Qi, Hongsheng Li, Zhanyu Ma, and Peng Gao. I-Max: Maximize the resolution potential of pre-trained rectified flow transformers with projected flow, 2024. arXiv:2410.07536.

[16] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, Dustin Podell, Tim Dockhorn, Zion English, and Robin Rombach. Scaling rectified flow transformers for high-resolution image synthesis. In *International Conference on Machine Learning (ICML)*, 2024.

[17] Fabian Falck et al. A Fourier space perspective on diffusion models, 2025. arXiv:2505.11278.

[18] David J. Field. Relations between the statistics of natural images and the response properties of cortical cells. *Journal of the Optical Society of America A*, 4(12):2379–2394, 1987.

[19] Malgorzata Galinska, Bart Pogodzinski, and Jan Eric Lenssen. CSFlow: Aligning flow matching with human contrast sensitivity, 2026. URL https://arxiv.org/abs/2606.08833.

[20] M. A. Georgeson and G. D. Sullivan. Contrast constancy: Deblurring in human vision by spatial frequency channels. *The Journal of Physiology*, 252(3):627–656, 1975.

[21] Alexandros Graikos, Nebojsa Jojic, and Dimitris Samaras. On the spectral properties of generative denoiser jacobians. *arXiv preprint arXiv:2609.36210*, 2026. URL https://arxiv.org/abs/2609.36210.

[22] Yingqing He, Shaoshu Yang, Haoxin Chen, Xiaodong Cun, Menghan Xia, Yong Zhang, Xintao Wang, Ran He, Qifeng Chen, and Ying Shan. ScaleCrafter: Tuning-free higher-resolution visual generation with diffusion models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.07702.

[23] David J. Heeger. Normalization of cell responses in cat striate cortex. *Visual Neuroscience*, 9(2):181–197, 1992.

[24] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. CLIPScore: A reference-free evaluation metric for image captioning. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

[25] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[26] Linjiang Huang, Rongyao Fang, Aiping Zhang, Guanglu Song, Si Liu, Yu Liu, and Hongsheng Li. FouriScale: A frequency perspective on training-free high-resolution image synthesis. In *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2403.12963.

[27] Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, and Raanan Fattal. DyPE: Dynamic position extrapolation for ultra high resolution diffusion, 2025. arXiv:2510.20766.

[28] Zhiyu Jin, Xuli Shen, Bin Li, and Xiangyang Xue. Training-free diffusion model adaptation for variable-sized text-to-image synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.08645.

[29] Zahra Kadkhodaie, Florentin Guth, Eero P. Simoncelli, and Stéphane Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.02557.

[30] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. MUSIQ: Multi-scale image quality transformer. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[31] Gihoon Kim and Taesup Kim. Correcting guided diffusion trajectories with spectral alignment, 2026. URL https://arxiv.org/abs/2610.02753.

[32] Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-pic: An open dataset of user preferences for text-to-image generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[33] David C. Knill, David Field, and Daniel Kersten. Human discrimination of fractal images. *Journal of the Optical Society of America A*, 7(6):1113–1123, 1990.

[34] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.02747.

[35] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.03003.

[36] Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, and Chengming Zhang. TIDE: Text-informed dynamic extrapolation with step-aware temperature control for diffusion transformers, 2026. URL https://arxiv.org/abs/2603.08928.

[37] Ken M. Nakanishi. Scalable-softmax is superior for attention, 2025. arXiv:2501.19399.

[38] Seunghun Oh and Unsang Park. Attention frequency modulation: Training-free spectral modulation of diffusion cross-attention. *arXiv preprint arXiv:2603.28114*, 2026. URL https://arxiv.org/abs/2603.28114.

[39] C. A. Párraga, T. Troscianko, and D. J. Tolhurst. The human visual system is optimised for processing the spatial information in natural visual images. *Current Biology*, 10(1):35–38, 2000.

[40] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[41] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.00071.

[42] Haonan Qiu, Shiwei Zhang, Yujie Wei, Ruihang Chu, Hangjie Yuan, Xiang Wang, Yingya Zhang, and Ziwei Liu. FreeScale: Unleashing the resolution of diffusion models via tuning-free scale fusion, 2024. arXiv:2412.09626.

[43] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

[44] Javad Rajabi, Koorosh Roohi, David B. Lindell, and Babak Taati. AReA: Adaptive receptive-field attention for efficient high-resolution diffusion transformers, 2026. URL https://arxiv.org/abs/2610.17227.

[45] Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers, 2026. URL https://arxiv.org/abs/2605.22668. arXiv:2605.22668.

[46] Odelia Schwartz and Eero P. Simoncelli. Natural signal statistics and sensory gain control. *Nature Neuroscience*, 4(8):819–825, 2001.

[47] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024. arXiv:2104.09864.

[48] J. H. van Hateren. A theory of maximizing sensory information. *Biological Cybernetics*, 68(1):23–29, 1992.

[49] Patrick von Platen, Suraj Patil, Anton Lozhkov, Pedro Cuenca, Nathan Lambert, Kashif Rasul, Mishig Davaadorj, Dhruv Nair, Sayak Paul, William Berman, Yiyi Xu, Steven Liu, and Thomas Wolf. Diffusers: State-of-the-art diffusion models. https://github.com/huggingface/diffusers, 2022.

[50] Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. In *AAAI Conference on Artificial Intelligence*, 2023.

[51] Michael A. Webster, Mark A. Georgeson, and Shernaaz M. Webster. Neural adjustments to image blur. *Nature Neuroscience*, 5(9):839–840, 2002.

[52] Jing Xiong, Liyang Fan, Hui Shen, Zunhai Su, Min Yang, Lingpeng Kong, and Ngai Wong. DoPE: Denoising rotary position embedding, 2025. URL https://arxiv.org/abs/2511.09146v1.

[53] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and evaluating human preferences for text-to-image generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[54] Yu Xue, Haoxuan Qu, Zhuoling Li, Hongbin Xu, Jianxiong Yin, Simon See, Hossein Rahmani, and Jun Liu. HRDiT: Training-free high-resolution image generation with off-the-shelf diffusion transformer models, 2026. URL https://arxiv.org/abs/2608.07003.

[55] Jinjin Zhang, Qiuyu Huang, Junjie Liu, Xiefan Guo, and Di Huang. Diffusion-4K: Ultra-high-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[56] Shen Zhang, Zhaowei Chen, Zhenyu Zhao, Yuhao Chen, Yao Tang, and Jiajun Liang. HiDiffusion: Unlocking higher-resolution creativity and efficiency in pretrained diffusion models. In *European Conference on Computer Vision (ECCV)*, 2024. arXiv:2311.17528.

[57] Zhengqiang Zhang, Ruihuang Li, and Lei Zhang. FreCaS: Efficient higher-resolution image generation via frequency-aware cascaded sampling. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.18410.

[58] Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.15894.

[59] Min Zhao, Bokai Yan, Xue Yang, Hongzhou Zhu, Jintao Zhang, Shilong Liu, Chongxuan Li, and Jun Zhu. UltraImage: Rethinking resolution extrapolation in image diffusion transformers, 2025. URL https://arxiv.org/abs/2512.04504.

A Additional figures

Figure A.1 shows held-out samples, Figure A.2 the dev pilot of SEGA's paper and code, and Figure A.3 the octave-band spectra discussed in Section 5.5.

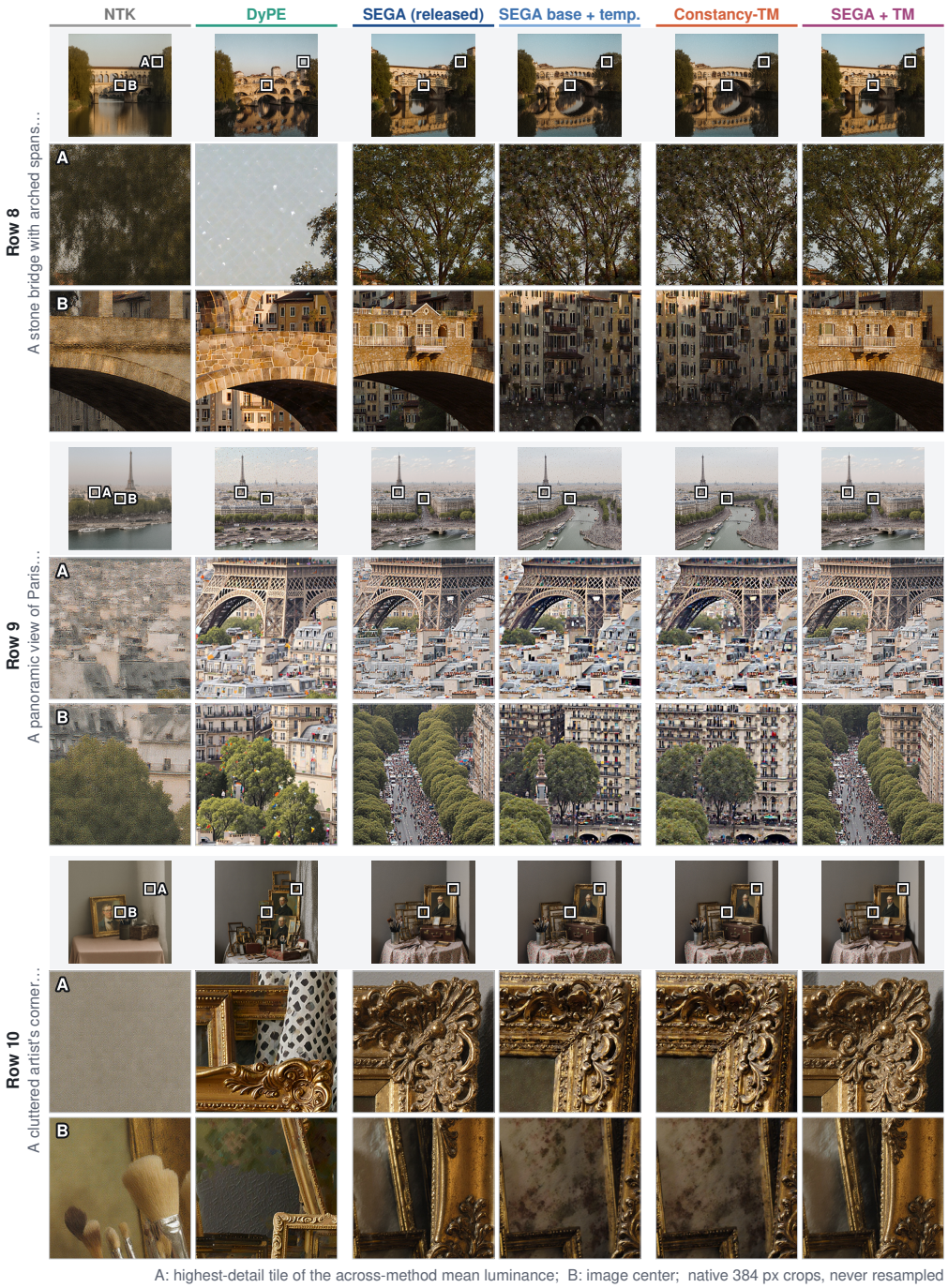


Figure A.1: At 4096^2 NTK loses fine detail, while the other arms render comparable detail. Held-out rows 8 to 10, the first three held-out rows, fixed before any output was viewed; seed 0, one sample per arm. Full frames are Lanczos-reduced; native-resolution crops sit at identical coordinates (A: highest-detail tile of the across-arm mean luminance; B: image centre) and are never resampled. Compositions differ between arms, so a crop can show different content. DyPE shows faint speckle in flat sky on held-out row 8 (one crop checked).

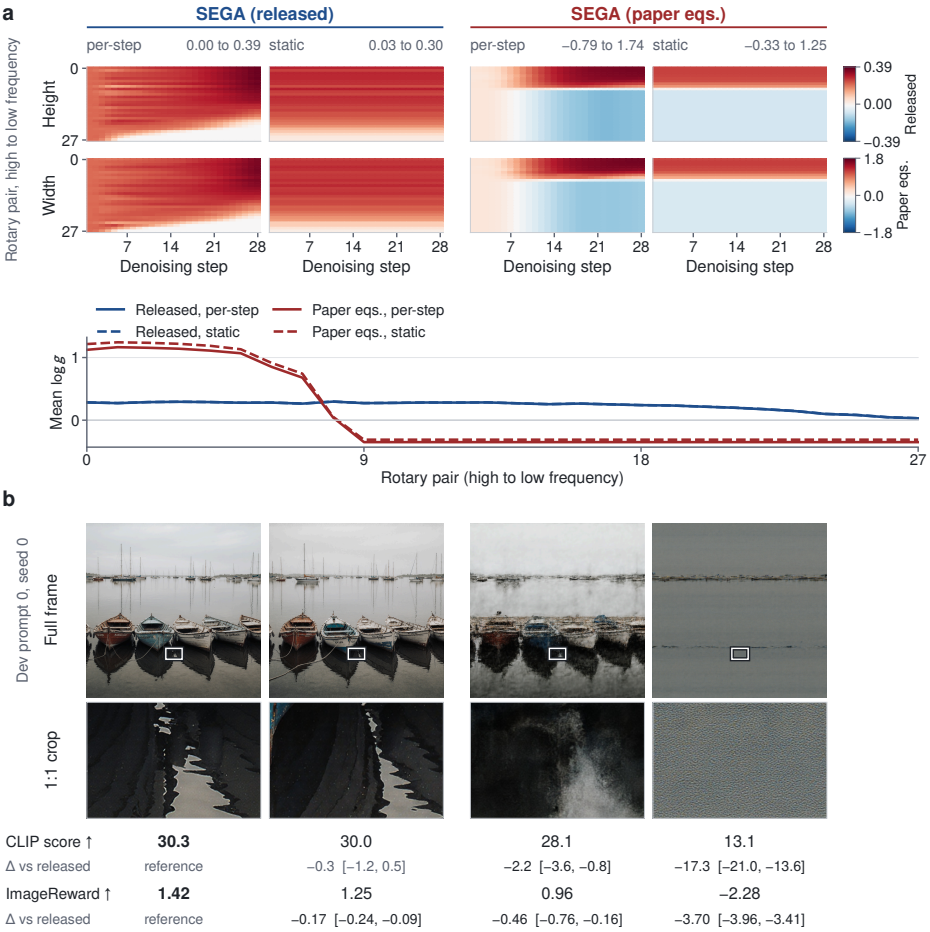


Figure A.2: The printed-law implementation produces larger gain changes than the release and degrades this development pilot. Eight dev prompts (rows 0–7, seed 0), descriptive only. (a) Median log logit gain per rotary pair and denoising step. Released and printed-law heatmaps have separate, explicitly labelled symmetric colour scales, so equal colours across those groups do not mean equal magnitudes. The line panel below uses one shared axis for mean log gain over prompts, steps and the two spatial axes for all four variants; static variants average each method's gains over prompts and steps. (b) Dev prompt 0, seed 0: full frames and a native-resolution crop, above CLIP score and ImageReward means over the eight dev prompts with paired differences from SEGA (released) and 95% bootstrap intervals.

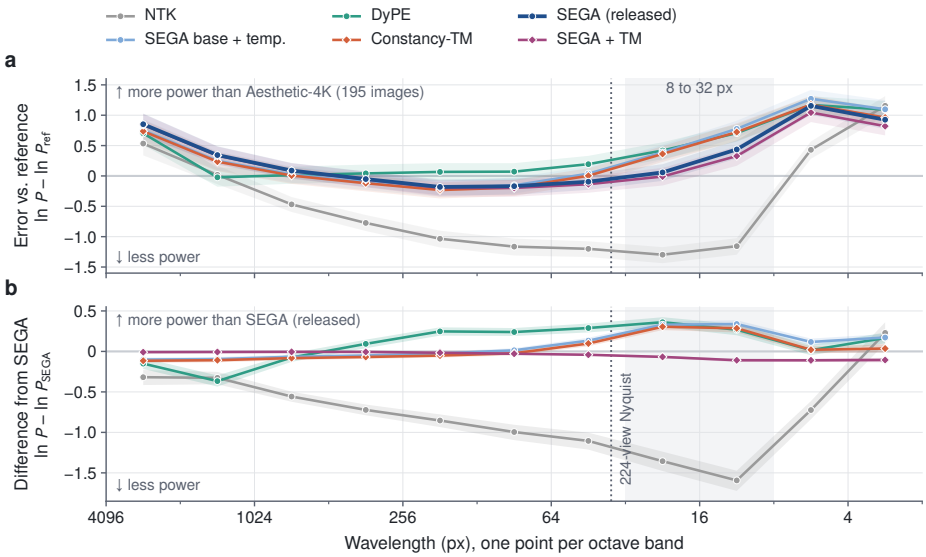


Figure A.3: Without SEGA's per-band allocation, luminance power at 8 to 32 pixel wavelengths sits above released SEGA's; Constancy-TM leaves a similar excess, and NTK falls short from 8 to 1024 pixels. (a) Signed log-power error against the mean ln P of the 195 Aesthetic-4K references, per octave band of linear-sRGB luminance; (b) paired difference from SEGA (released). Held-out rows 8–194 (n = 187, seed 0); bands are 95% bootstrap intervals over prompts. The shaded 8 to 32 pixel band lies above the Nyquist frequency of a 224-pixel view.

B Full held-out results

Table B.1: Base and temperature carry most of SEGA's gain, and no per-band law beats it. Held-out paired differences against SEGA (released) (rows 8–194, seed 0, $n = 187$); the first row gives the absolute means of SEGA (released). Each cell shows the mean difference and its nominal 95% bootstrap interval; ▲/▼ and colour mark intervals that exclude zero in the better or worse direction, while uncoloured cells are ties. A star marks a Holm-adjusted $p \leq 0.05$ across this table's paired comparisons, using approximate centered-bootstrap tests. Table B.4 has all ten metrics and a separate Holm family.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑
SEGA (released)	1.190	23.05	29.14	45.23	58.33	0.603
Decomposition of SEGA (released)						
NTK	−0.594* ▼ [−0.701, −0.494]	−1.02* ▼ [−1.16, −0.89]	−1.99* ▼ [−2.43, −1.55]	−15.74* ▼ [−17.50, −14.00]	−18.70* ▼ [−20.68, −16.86]	−0.046* ▼ [−0.062, −0.031]
SEGA base + temp.	+0.026 [−0.016, +0.070]	+0.03 [−0.03, +0.09]	+0.02 [−0.19, +0.23]	−2.09* ▼ [−3.25, −0.96]	−0.11 [−0.71, +0.51]	−0.009 ▼ [−0.018, −0.001]
Other baseline						
DyPE	−0.164* ▼ [−0.258, −0.087]	−0.49* ▼ [−0.63, −0.36]	−0.37 ▼ [−0.72, −0.01]	−1.88 ▼ [−3.26, −0.56]	+1.97* ▲ [+1.16, +2.80]	+0.029* ▲ [+0.020, +0.038]
Principled per-band laws (ours)						
Constancy-TM	+0.018 [−0.023, +0.059]	−0.05 [−0.11, +0.02]	−0.27 ▼ [−0.49, −0.04]	−3.04* ▼ [−4.28, −1.81]	−2.19* ▼ [−2.81, −1.57]	−0.025* ▼ [−0.034, −0.017]
SEGA + TM	+0.009 [−0.008, +0.027]	−0.08* ▼ [−0.10, −0.05]	−0.09 ▼ [−0.20, −0.00]	−0.21 [−0.71, +0.32]	−1.13* ▼ [−1.39, −0.90]	+0.003 [−0.000, +0.006]

Table B.2 decomposes SEGA's gain over NTK with means and paired gains, Table B.3 gives absolute means and Table B.4 the paired differences for all ten metrics. The spectral rows use the mean octave profile of the 195 Aesthetic-4K reference images as the target; Table B.5 breaks the spectral error into its 11 octave bands. FID and KID are not tabulated: for the seed-1 images (56 prompts per arm, including one dev row), the paired FID difference of SEGA base + temp. against SEGA (released) is +3.26 with interval [−6.86, +13.87], and that of Constancy-TM is +3.74 with interval [−6.72, +14.08].

Table B.2: Where SEGA's improvement over NTK comes from. Means on the same 187 held-out prompts, paired gains from base and temperature and from the allocation, and the share of the total gain carried by base and temperature, (SEGA base + temp. − NTK)/(SEGA (released) − NTK), with its 95% paired bootstrap interval. A share above 100% means the allocation's point estimate is negative.

Metric	Mean ($n = 187$)			Paired gain		
	NTK	SEGA base + temp.	SEGA (released)	base + temp.	alloc-ation	share of base + temp.
ImageReward↑	0.596	1.216	1.190	+0.620 ▲ [+0.525, +0.724]	−0.026 [−0.070, +0.016]	104% [97, 113]
PickScore↑	22.03	23.08	23.05	+1.05 ▲ [+0.92, +1.18]	−0.03 [−0.09, +0.03]	103% [97, 109]
CLIP↑	27.15	29.15	29.14	+2.01 ▲ [+1.57, +2.48]	−0.02 [−0.23, +0.19]	101% [90, 112]
MUSIQ↑	29.49	43.14	45.23	+13.65 ▲ [+12.05, +15.28]	+2.09 ▲ [+0.96, +3.25]	87% [80, 94]
MUSIQcrop↑	39.63	58.22	58.33	+18.59 ▲ [+16.72, +20.63]	+0.11 [−0.51, +0.71]	99% [96, 103]
CLIP-IQA↑	0.557	0.594	0.603	+0.037 ▲ [+0.021, +0.053]	+0.009 ▲ [+0.001, +0.018]	80% [59, 98]

Table B.3: Held-out means with 95% bootstrap intervals (rows 8–194, seed 0). Constancy is the partial run on 41 prompts (Appendix D).

Metric	SEGA (released) $n = 187$	NTK $n = 187$	SEGA base + temp. $n = 187$	DyPE $n = 187$	Constancy $n = 41$	Constancy-TM $n = 187$	SEGA + TM $n = 187$
ImageReward↑	1.190 [1.087, 1.284]	0.596 [0.456, 0.736]	1.216 [1.118, 1.313]	1.026 [0.910, 1.142]	1.177 [0.953, 1.392]	1.208 [1.110, 1.304]	1.199 [1.096, 1.293]
PickScore↑	23.05 [22.89, 23.21]	22.03 [21.84, 22.21]	23.08 [22.92, 23.24]	22.57 [22.36, 22.75]	22.70 [22.29, 23.08]	23.01 [22.84, 23.16]	22.98 [22.81, 23.13]
CLIP↑	29.14 [28.70, 29.55]	27.15 [26.64, 27.64]	29.15 [28.72, 29.58]	28.76 [28.32, 29.20]	29.46 [28.40, 30.49]	28.87 [28.42, 29.31]	29.04 [28.61, 29.47]
MUSIQ↑	45.23 [43.69, 46.79]	29.49 [28.13, 30.95]	43.14 [41.58, 44.61]	43.35 [41.71, 45.02]	40.55 [37.20, 43.56]	42.19 [40.65, 43.58]	45.02 [43.56, 46.53]
MUSIQcrop↑	58.33 [57.06, 59.63]	39.63 [37.77, 41.33]	58.22 [57.07, 59.44]	60.30 [59.12, 61.53]	60.75 [58.19, 63.01]	56.14 [54.98, 57.36]	57.20 [55.84, 58.52]
CLIP-IQA↑	0.603 [0.590, 0.616]	0.557 [0.546, 0.568]	0.594 [0.581, 0.607]	0.632 [0.618, 0.645]	0.572 [0.544, 0.601]	0.578 [0.566, 0.590]	0.606 [0.593, 0.619]
CLIP-IQAcrop↑	0.586 [0.573, 0.600]	0.537 [0.525, 0.549]	0.569 [0.556, 0.583]	0.599 [0.585, 0.613]	0.543 [0.514, 0.572]	0.550 [0.538, 0.563]	0.588 [0.576, 0.602]
Spec. err.↓	1.068 [1.005, 1.133]	1.285 [1.214, 1.358]	1.102 [1.035, 1.168]	1.096 [1.024, 1.168]	1.337 [1.205, 1.477]	1.088 [1.020, 1.152]	1.066 [1.001, 1.133]
Slope err.↓	0.200 [0.180, 0.221]	0.165 [0.146, 0.185]	0.224 [0.200, 0.247]	0.218 [0.194, 0.242]	0.276 [0.218, 0.338]	0.218 [0.195, 0.240]	0.197 [0.178, 0.218]
Tiling↓	0.0684 [0.0568, 0.0802]	0.0757 [0.0612, 0.0905]	0.0616 [0.0507, 0.0732]	0.0865 [0.0778, 0.0953]	0.0589 [0.0390, 0.0797]	0.0616 [0.0505, 0.0732]	0.0684 [0.0566, 0.0804]

Table B.4: Held-out paired differences against SEGA (released), all metrics. Nominal 95% intervals and colour as in Table B.1; stars mark approximate Holm-adjusted $p \leq 0.05$ across all paired comparisons in this table. Constancy is the partial run on 41 prompts (Appendix D).

Metric	NTK $n = 187$	SEGA base + temp. $n = 187$	DyPE $n = 187$	Constancy $n = 41$	Constancy-TM $n = 187$	SEGA + TM $n = 187$
ImageReward↑	−0.594* ▼ [−0.701, −0.494]	+0.026 [−0.016, +0.070]	−0.164* ▼ [−0.258, −0.087]	+0.018 [−0.089, +0.133]	+0.018 [−0.023, +0.059]	+0.009 [−0.008, +0.027]
PickScore↑	−1.02* ▼ [−1.16, −0.89]	+0.03 [−0.03, +0.09]	−0.49* ▼ [−0.63, −0.36]	−0.07 [−0.28, +0.18]	−0.05 [−0.11, +0.02]	−0.08* ▼ [−0.10, −0.05]
CLIP↑	−1.99* ▼ [−2.43, −1.55]	+0.02 [−0.19, +0.23]	−0.37 ▼ [−0.72, −0.01]	−0.05 [−0.78, +0.68]	−0.27 ▼ [−0.49, −0.04]	−0.09 ▼ [−0.20, −0.00]
MUSIQ↑	−15.74* ▼ [−17.50, −14.00]	−2.09* ▼ [−3.25, −0.96]	−1.88 ▼ [−3.26, −0.56]	−5.09 ▼ [−8.44, −1.74]	−3.04* ▼ [−4.28, −1.81]	−0.21 [−0.71, +0.32]
MUSIQcrop↑	−18.70* ▼ [−20.68, −16.86]	−0.11 [−0.71, +0.51]	+1.97* ▲ [+1.16, +2.80]	+1.91 ▲ [+0.10, +3.80]	−2.19* ▼ [−2.81, −1.57]	−1.13* ▼ [−1.39, −0.90]
CLIP-IQA↑	−0.046* ▼ [−0.062, −0.031]	−0.009 ▼ [−0.018, −0.001]	+0.029* ▲ [+0.020, +0.038]	−0.005 [−0.031, +0.019]	−0.025* ▼ [−0.034, −0.017]	+0.003 [−0.000, +0.006]
CLIP-IQAcrop↑	−0.049* ▼ [−0.066, −0.034]	−0.017* ▼ [−0.026, −0.008]	+0.013 ▲ [+0.003, +0.023]	−0.027 [−0.053, +0.001]	−0.036* ▼ [−0.045, −0.027]	+0.002 [−0.001, +0.006]
Spec. err.↓	+0.216* ▼ [+0.154, +0.287]	+0.034* ▼ [+0.016, +0.051]	+0.028 [−0.007, +0.061]	+0.204* ▼ [+0.127, +0.277]	+0.019 ▼ [+0.001, +0.037]	−0.002 [−0.009, +0.004]
Slope err.↓	−0.035* ▲ [−0.053, −0.018]	+0.024* ▼ [+0.013, +0.034]	+0.018 ▼ [+0.003, +0.033]	+0.061 ▼ [+0.021, +0.101]	+0.018* ▼ [+0.008, +0.027]	−0.003 [−0.007, +0.002]
Tiling↓	+0.0074 [−0.0044, +0.0185]	−0.0068* ▲ [−0.0108, −0.0026]	+0.0182* ▼ [+0.0085, +0.0279]	−0.0099 [−0.0223, +0.0041]	−0.0068* ▲ [−0.0107, −0.0026]	+0.0001 [−0.0007, +0.0008]

Table B.5: Spectral error by octave band. Mean log-power difference from the Aesthetic-4K reference profile on held-out prompts; positive values mean excess power. Bands 7 to 10 lie above the Nyquist frequency of a 224-pixel view of a 4096^2 image. Constancy is the partial run on 41 prompts (Appendix D).

Octave band	0	1	2	3	4	5	6	7	8	9	10
Centre (cyc/px)	$2^{-11.5}$	$2^{-10.5}$	$2^{-9.5}$	$2^{-8.5}$	$2^{-7.5}$	$2^{-6.5}$	$2^{-5.5}$	$2^{-4.5}$	$2^{-3.5}$	$2^{-2.5}$	$2^{-1.5}$
SEGA (released)	+0.85	+0.34	+0.09	−0.05	−0.18	−0.17	−0.10	+0.06	+0.44	+1.15	+0.93
NTK	+0.53	+0.02	−0.47	−0.77	−1.03	−1.16	−1.20	−1.30	−1.16	+0.43	+1.15
SEGA base + temp.	+0.75	+0.25	+0.02	−0.10	−0.21	−0.15	+0.04	+0.39	+0.78	+1.27	+1.10
DyPE	+0.70	−0.02	+0.01	+0.04	+0.07	+0.07	+0.19	+0.42	+0.71	+1.17	+1.09
Constancy	+0.73	+0.34	+0.27	+0.23	+0.21	+0.36	+0.72	+1.37	+1.72	+1.84	+1.64
Constancy-TM	+0.74	+0.24	+0.01	−0.12	−0.23	−0.19	+0.00	+0.36	+0.72	+1.17	+0.96
SEGA + TM	+0.84	+0.34	+0.08	−0.06	−0.20	−0.20	−0.14	−0.01	+0.33	+1.04	+0.82

C Seed-1 replication

Table C.1 repeats the comparison for SEGA base + temp. and Constancy-TM on held-out rows 8–62 with a second seed, next to seed 0 on the same 55 prompts. One of the verdicts that changes, crop CLIP-IQA for SEGA base + temp. at seed 1, rests on an interval that excludes zero by less than 10^{-4}, so the agreement count can move by one under another bootstrap seed. Every verdict that changes moves between a tie and a one-sided verdict. Spectral log-RMSE differences grow at seed 1 (SEGA base + temp.: +0.080 against +0.056 at seed 0 on the same prompts; Constancy-TM: +0.062 against +0.042).

Table C.1: Seed 1 repeats most nominal seed-0 verdicts. Paired differences against SEGA (released) on the same 55 held-out prompts (rows 8–62) at seed 0 and at seed 1. 17 of 20 nominal verdicts agree. Intervals, colour and triangles use the nominal 95% convention of Table B.1; this descriptive table has no Holm adjustment.

	SEGA base + temp. − SEGA (released)		Constancy-TM − SEGA (released)	
Metric	seed 0	seed 1	seed 0	seed 1
ImageReward↑	+0.070 [−0.011, +0.160]	+0.031 [−0.036, +0.094]	+0.063 [−0.014, +0.142]	+0.019 [−0.045, +0.083]
PickScore↑	+0.05 [−0.07, +0.20]	+0.00 [−0.12, +0.13]	−0.01 [−0.12, +0.12]	−0.08 [−0.20, +0.05]
CLIP↑	−0.02 [−0.54, +0.47]	+0.02 [−0.40, +0.46]	−0.31 [−0.82, +0.17]	−0.32 [−0.78, +0.13]
MUSIQ↑	−3.62 ▾ [−5.84, −1.52]	−2.79 ▾ [−4.58, −0.95]	−4.36 ▾ [−6.57, −2.32]	−3.53 ▾ [−5.38, −1.69]
MUSIQcrop↑	+0.43 [−0.53, +1.40]	+0.35 [−0.69, +1.44]	−1.77 ▾ [−2.82, −0.77]	−1.99 ▾ [−3.09, −0.91]
CLIP-IQA↑	+0.002 [−0.012, +0.017]	−0.008 [−0.023, +0.008]	−0.018 ▾ [−0.034, −0.002]	−0.025 ▾ [−0.043, −0.007]
CLIP-IQAcrop↑	−0.008 [−0.024, +0.009]	−0.015 ▾ [−0.032, −0.000]	−0.034 ▾ [−0.050, −0.015]	−0.042 ▾ [−0.059, −0.025]
Spec. err.↓	+0.056 ▾ [+0.020, +0.091]	+0.080 ▾ [+0.042, +0.120]	+0.042 ▾ [+0.006, +0.077]	+0.062 ▾ [+0.026, +0.101]
Slope err.↓	+0.035 ▾ [+0.018, +0.053]	+0.028 ▾ [+0.005, +0.050]	+0.029 ▾ [+0.013, +0.046]	+0.025 ▾ [+0.004, +0.046]
Tiling↓	−0.0097 ▴ [−0.0157, −0.0038]	−0.0055 [−0.0136, +0.0027]	−0.0098 ▴ [−0.0156, −0.0040]	−0.0058 [−0.0139, +0.0023]

D Partial and dev-only runs (not held-out results)

Nothing in this appendix is a held-out result of the main protocol. The first subsection reports a held-out run that covers only part of the held-out set and started before the protocol was fixed; the other two report dev-prompt selection evidence and a descriptive pilot.

D.1 Partial held-out run of Constancy

The original Constancy was generated on 41 of the 187 held-out prompts (seed 0) before the protocol was fixed (Appendix D.2). Every number in this subsection comes from those 41 prompts only, and differences are paired on them. Against SEGA (released) the run ties on the reward models, loses 5.09 full-frame MUSIQ points, gains crop MUSIQ (+1.91) and has a much larger spectral error (+0.204), with excess power in the top octaves and the largest excess of any arm in band 8 (log-power error +1.72; Table B.5). Measured as a share of the gain of SEGA (released) over NTK on the same prompts ($n = 41$), it recovers 110% [100, 120] on crop MUSIQ but 68% [47, 88] on full-frame MUSIQ.

The visibility gate of Section 5.2 opens late on these prompts as on the dev prompts. On the 41 held-out prompts it opens only after the first 20 of 28 steps (median; range 16 to 22), at $\sigma = 0.56$ (range 0.70 to 0.46), and the correction then acts in a median of 8 steps.

This partial run offers suggestive evidence about temperature. NTK and SEGA base + temp. differ in both base and temperature, while Constancy adds to the NTK base the log-N temperature ($\tau = 1.333$) and a per-band correction that stays at zero for the first 20 of 28 steps (median). On its 41 prompts it improves on NTK by +0.688 [+0.479, +0.917] ImageReward and +21.66 [+17.43, +26.06] crop MUSIQ; on the same prompts, the entire SEGA (released) minus NTK gap is +0.670 ImageReward and +19.75 crop MUSIQ. The rows are a nonrandom manifest prefix, the temperature differs from SEGA's, and the correction eventually acts. These observations do not identify the separate base and temperature contributions.

D.2 Dev selection

Table D.1 lists every variant scored on the eight dev prompts, with its mean and its difference from the SEGA (released) reference of the same campaign. These numbers chose what to run held-out; they are not results. Campaign 1 preceded the written gate: it sent SEGA base + temp. to the held-out set as a decomposition control and Constancy-TM as the candidate, and seven chunks of the original Constancy had already run on held-out rows when the protocol was fixed, which is why that arm has only 41 held-out prompts. Campaigns 2 and 3 applied the gate of Section 4. In campaign 3 a fresh SEGA (released) rerun served as the reference. It differs from the pilot run of the same prompts and seed by +0.79 MUSIQ and −0.032 ImageReward, which is more than the MUSIQ margin by which the one passing variant cleared the gate (+0.39); we did not establish the source of the difference. That variant was not run on held-out prompts, and we make no claim about it. The dev set also misjudged the allocation itself: on the eight dev prompts SEGA base + temp. scored +0.24 MUSIQ against SEGA (released), while held out it trails by 2.09.

Table D.1: Dev-set selection (rows 0–7, seed 0, $n = 8$); selection evidence only. Each cell: mean, and below it the difference from the campaign's SEGA (released) reference. [a]Partial held-out run (41 prompts) started before the dev decision. [b]Fresh rerun; the second line is its difference from the pilot run. [c]Passed the dev gate; never evaluated on held-out prompts.

Variant	Image-Reward↑	MUSIQ full↑	CLIP score↑	Pick-Score↑	MUSIQ crop↑	Outcome
Campaign 1 (reference: SEGA (released), pilot run)						
SEGA (released)	1.423 +0.000	42.49 +0.00	30.32 +0.00	23.41 +0.00	58.23 +0.00	reference
NTK + log-N temp.	1.282 −0.141	40.36 −2.13	29.96 −0.36	23.12 −0.29	60.29 +2.05	dev only
Constancy	1.285 −0.138	40.56 −1.93	30.10 −0.22	23.10 −0.31	58.62 +0.39	partial held-out[a]
Constancy-TM on NTK base, $\eta = 1$	1.314 −0.109	41.87 −0.62	30.04 −0.28	23.10 −0.31	57.51 −0.72	dev only
Constancy-TM on NTK base, $\eta = 2$	1.313 −0.110	43.34 +0.85	30.13 −0.19	23.04 −0.37	54.61 −3.63	dev only
SEGA base + temp.	1.444 +0.021	42.72 +0.24	30.32 +0.00	23.26 −0.15	55.93 −2.30	held-out (control)
Constancy-TM	1.432 +0.009	41.65 −0.84	30.24 −0.08	23.25 −0.16	54.57 −3.66	held-out (candidate)
Campaign 2, written gate (reference: SEGA (released), pilot run)						
Alias-T	1.281 −0.142	35.81 −6.68	29.37 −0.95	23.06 −0.35	63.66 +5.43	fails gate
Alias-T-s	1.277 −0.146	44.97 +2.48	30.24 −0.08	23.47 +0.06	58.73 +0.49	fails gate
SEGA + TM, $\eta = 0.5$	1.441 +0.018	42.48 −0.01	30.20 −0.12	23.31 −0.10	58.51 +0.27	fails gate
SEGA + TM, $\eta = 1$	1.449 +0.026	42.51 +0.02	30.34 +0.02	23.33 −0.08	58.28 +0.04	passes, held-out
Campaign 3, same gate (reference: SEGA (released), fresh rerun)						
SEGA (released) (rerun)	1.391 −0.032	43.28 +0.79	30.15 −0.17	23.38 −0.03	58.59 +0.36	reference[b]
SEGA-RMS	1.428 +0.036	43.67 +0.39	30.08 −0.07	23.31 −0.07	58.38 −0.21	passes, not confirmed[c]
SEGA-Uniform	1.374 −0.017	41.71 −1.57	30.73 +0.59	23.29 −0.08	58.07 −0.52	fails gate
SEGA + TM-lag	1.411 +0.019	43.06 −0.22	30.16 +0.01	23.32 −0.05	58.44 −0.15	fails gate

D.3 SEGA paper versus code: pilot numbers

In an eight-prompt pilot on the dev prompts, which is descriptive only, our implementation of the printed equations scores ImageReward 0.964, CLIP 28.14 and full-frame MUSIQ 30.78, against 1.423, 30.32 and 42.49 for the released code (Table D.2). Freezing the printed allocation into a static tilt, a natural ablation of content adaptivity, destroys the images: CLIP falls to 13.06 and logit gains reach 3.49. Freezing the released allocation instead costs much less (CLIP 29.98, ImageReward 1.255). Figure A.2 in Appendix A shows the two gain trajectories and the resulting images. The pilot also ran YaRN, which follows a literal reading of a design note rather than the usual rotation-count convention.

Table D.2: Eight-prompt pilot on the dev prompts (rows 0–7, seed 0), descriptive only. Means; colour marks a paired bootstrap interval against SEGA (released) that excludes zero, which with $n = 8$ is indicative at best. Static variants freeze the allocation instead of reading the current latent.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑	Spectral error↓
SEGA (released)	1.423	23.41	30.32	42.49	58.23	0.557	0.857
SEGA (released, static)	1.255 ▾	23.12	29.98	43.25	57.75	0.559	0.870
SEGA (paper eqs.)	0.964 ▾	22.18 ▾	28.14 ▾	30.78 ▾	42.73 ▾	0.455 ▾	0.944
SEGA (paper eqs., static)	−2.277 ▾	16.93 ▾	13.06 ▾	28.49 ▾	35.53 ▾	0.435 ▾	4.369 ▾
NTK	0.757 ▾	22.37 ▾	27.84 ▾	32.91 ▾	41.41 ▾	0.543	1.220 ▾
YaRN	−0.163 ▾	21.14 ▾	28.06 ▾	42.51	66.69 ▴	0.645 ▴	1.250 ▾
DyPE	1.118 ▾	22.23 ▾	28.59 ▾	44.38	58.32	0.584	0.920
NTK + log-N temp.	1.282 ▾	23.12	29.96	40.36	60.29	0.578	1.038 ▾
Constancy	1.285 ▾	23.10	30.10	40.56	58.62	0.563	1.017 ▾

E Implementation details

Pipeline. The FLUX.1-Krea-dev transformer, loaded through an environment override (`CROPE_TRANSFORMER`), runs inside the FLUX.1-dev pipeline (T5 and CLIP text encoders with 512 T5 tokens, autoencoder with tiled decoding at 4096^2), diffusers 0.39 [49], PyTorch 2.11, bfloat16, batch size one, a per-prompt CPU seed, 28 Euler steps and distilled guidance 4.5. The transformer has 19 double-stream and 38 single-stream blocks with 24 heads of dimension 128, and normalizes q and k before the rotation. Every arm multiplies q and k of every token in every block by one gain vector after the rotation; the vector is shared by blocks and heads and recomputed once per model call. The scheduler patch that holds $\mu = 1.15$ is applied before any configuration is recorded. The per-image JSON records hard-code the base model name and an attention label; the loaded transformer comes from the task scripts and the attention policy from the code revision (Section 5.5 and Appendix G).

MUSIQ input pipeline. We use pyiqa 0.1.16's KonIQ10k MUSIQ model. The project passes RGB float32 values in $[0, 1]$ without resizing; MUSIQ normalizes to $[-1, 1]$ and combines the native image with aspect-preserving 224- and 384-pixel views, resized by bicubic interpolation with `align_corners=False` and `antialias=False`. Each scale is split into 32×32 patches at stride 32, with SAME padding where needed. The native setting `max_seq_len_from_original_res=-1` retains all patches: a 4096^2 frame contributes 16,384 native patches, plus 49 and 144 resized-view patches, for 16,577 before the class token. A 1024^2 crop contributes 1,217. Neither native square grid needs padding. The spatial embedding's 10×10 hash grid does not downsample the image. We verified these counts by executing the installed preprocessor without loading model weights. Crop MUSIQ averages four separately scored, fixed random 1024^2 crops shared across methods. Full-frame MUSIQ is thus a multiscale quality score that includes every native pixel, not a direct sharpness measurement.

Source versions. Our SEGA comparison uses released commit `ddaab08552c7`. The allocation is in `flux_sega/transformer_flux.py`, lines 103–154; projected-feature sensing follows the input projection in the same file. Our printed-law implementation follows Section 4.1 and equations (3)–(7) of Rajabi et al. [45], with the strengthened base from equation (9) in its Appendix A.2.2. The six-row comparison in Table 2 records our interpretation, including how the unspecified index axis and latent input are implemented. We did not obtain confirmation of that interpretation from the authors. Our DyPE port and official-code diagnostic use commit `436fe612fb76`.

DyPE. For spatial axis a with scale s and $\kappa = \sigma^2$, let $b(r) = D \ln\big(64/(2\pi r)\big)/(2 \ln \theta)$ be the pair index that completes r rotations over 64 tokens and $\mu_{r_1,r_2}(j) = 1 - \mathrm{clip}\big((j - \ell)/(h - \ell), 0, 1\big)$ with $\ell = \max(\lfloor b(r_1) \rfloor, 0)$ and $h = \min(\lceil b(r_2) \rceil, D/2)$. Then

$$\omega_j^{\mathrm{mix}} = \tfrac{\omega_j}{s}\big(1 - \mu_{1.25\kappa,0.75\kappa}(j)\big) + \omega_j^{\mathrm{ntk}}\, \mu_{1.25\kappa,0.75\kappa}(j), \qquad \omega'_j = \omega_j^{\mathrm{mix}}\big(1 - \mu_{16\kappa,2\kappa}(j)\big) + \omega_j\, \mu_{16\kappa,2\kappa}(j),$$

with ω^{ntk} the NTK-aware frequencies, and the spatial cos/sin amplitude is $1 + 0.1 \ln s$, as in the official code at the inspected revision.

SEGA (released). The sensor takes the channel mean Z of the projected input features on the token grid. The axis profile E_a is the one-dimensional FFT power along axis a averaged over the other

axis and rebinned into $K_a = \max(\lfloor L_a/2 \rfloor, 8)$ bins. Pair j reads $\ell_j = \ln E_a\big((1-\rho_j)(K_a-1)\big)$ by linear interpolation, with $\rho_j = j/(D/2-1)$ the normalized rank of its log period; z_j standardizes ℓ_j with the sample standard deviation; and $m_{a,j} = \max\big(1, m_{\text{ref}}[1 - 0.15\,\varsigma\,c_j]\big)$ with $c_j = \tanh(1.5 z_j) - \text{mean}_i \tanh(1.5 z_i)$, $m_{\text{ref}} = \max(s,1)^{0.08}$ and $\varsigma = 1 - \text{SF}^{1.5}$ from the radially binned two-dimensional power. The index axis keeps gain 1.

Constancy. Band energies are densities: with an orthonormal FFT, $P(k) = \sum_c |\mathcal{F}(x_c - \bar{x}_c)(k)|^2$, $E_a(k)$ averages P over the other axis, and $E_a[b]$ averages over half-octave bands b (13 bands at 256 tokens per side). Unit white noise gives $E_a[b] = C = 64$ in expectation, so the noise floor at level σ is $\sigma^2 C$. The gate uses the current input; the error uses the clean estimate of the previous call, $\hat{x} = x_{n-1} - \sigma_{n-1} v_\theta(x_{n-1})$. Corrections are interpolated in $\log f$ between band centres and applied to pairs with frequency above one cycle per grid, which at 4096^2 are pairs 0 to 8 of each axis. Step 0 has no clean estimate and uses the temperature alone, and $|\ln g - \ln \tau| \le \eta$ always. The native targets come from 1024^2 samples of the same model on 16 Pick-a-Pic prompts [32] disjoint from the evaluation set: the late target averages the three steps with $\sigma < 0.3$, and the time-matched target keeps all 28 steps and is interpolated in σ.

Alias-T. Alias-T applies the log-N temperature of Section 2 pair by pair. A pair of wavelength $\lambda_{a,j}$ tokens under the released base, on a grid of L tokens per side ($L = 64$ at native size), gets the logit gain

$$g_{a,j} = \frac{\ln\big(n(\lambda_{a,j}, L)\, L\big)}{\ln\big(n(\lambda_{a,j}, 64)\cdot 64\big)}, \qquad n(\lambda, L) = \min\big(L, \max(1, L/\lambda)\big),$$

where $n(\lambda, L)$, the pair's alias count, is the number of its wavelengths across the grid, bounded by 1 and L; the index axis takes τ_N. The gains are static and content-free, with no correction and no further temperature. Alias-T-s shifts every $\ln g$ by one constant so that the mean over all 128 dimensions equals SEGA's reference log gain, $0.16 \ln s$.

F Mechanism diagnostics

CPU study of the printed SEGA law. Before running any model we measured the printed allocation (Equation (2)) on forward-noised latents of the FLUX.2 autoencoder for 20 DIV2K crops. A fixed tilt over pairs explained 71% of the variance of the modulation, tilt plus timestep 91%, leaving at most 9.2% for content; the flatness gate ς was explained to $R^2 = 0.83$ by the timestep alone and to 0.78 by a generic $1/f$ field. On our held-out FLUX.1-Krea-dev runs the released gate ς is likewise explained to $R^2 = 0.89$ by the timestep. The channel mean that the printed sensor reads keeps 1.7% of the latent power. In a blur sweep at $\sigma = 0.15$ over 10 test crops, the printed law's contrast between fine and coarse pairs changed by 12% between no blur and a blur of 2 tokens, while an oracle version of the constancy law changed 2.4-fold. These measurements motivated the constancy design; they use the printed law, another autoencoder and forward-noised rather than sampled latents, so we use them only as context for Section 5.2.

Probe protocol. The transfer probe of Figure F.1 reruns a dev prompt with the gain of one input octave multiplied by 1.5 on both spatial axes, with the same seed, and reports the finite-difference elasticity $\partial \log P_{\text{out}} / \partial \log g_{\text{in}}$ for six input and seven output octaves of the final latent and image. It used the NTK base with the log-N temperature, three dev prompts and one seed. The local-response experiment applied half-step perturbations at steps 6, 14 and 22 of two dev prompts and compared the observed change in band power with the prediction of a locally fitted linear response: the median error relative to predicting no change was 0.94 (range 0.71 to 1.35; 9 of 12 cases below 1). It tested a single octave, below the resolution of the token grid, so it does not cover all rotary bands. On the width axis of FLUX.1-Krea-dev, the mean diagonal elasticity of the transfer probe is 0.47 for one prompt but -0.03 and -0.04 for the other two, against a mean off-diagonal response of 0.09.

Confounded cross-checkpoint probe. Raising the fastest octave's gain gave a width-axis elasticity of 0.11 on FLUX.1-Krea-dev and -3.64 on FLUX.1-dev for one dev prompt. Guidance and code revision also differed, so we draw no conclusion about checkpoint-dependent response from the sign change.

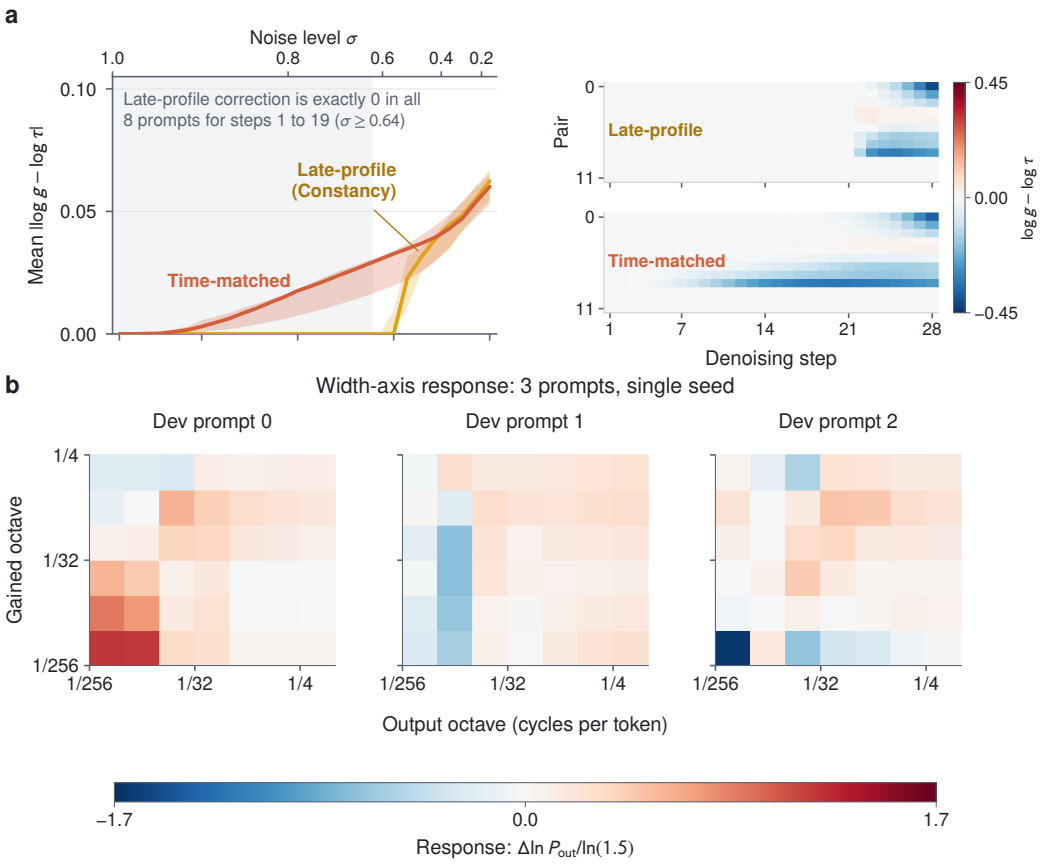


Figure F.1: The late-profile correction starts late, and the response to a band-gain change differs across prompts. (a) Mean $|\ln g - \ln \tau|$ over spatial rotary pairs per step, shown as the median and interquartile band over eight dev prompts, seed 0, with both targets on the NTK base; right, median corrections for pairs 0–11; omitted pairs 12–27 are identically zero. (b) Separate width-axis response matrices for dev prompts 0, 1 and 2, all at seed 0. Each cell is the same-seed finite-difference change in final-latent log power divided by $\ln(1.5)$ when one input octave receives a 1.5× logit gain on both spatial axes throughout sampling. All three matrices share one zero-centred colour scale and show no averaging across prompts; there is no uncertainty estimate from repeated seeds.

Allocation controls on the dev set. A third dev campaign replaced the allocation of SEGA (released) by a uniform gain with the same root-mean-square value, or kept its shape and removed the rise in mean gain caused by the clamp. On the eight dev prompts the uniform control changed MUSIQ by −1.57 and the shape-only control by +0.39, while a rerun of SEGA (released) itself moved MUSIQ by +0.79. These differences are within run-to-run variation and were not taken to held-out prompts.

Diagnostics not reported. The project also measured run-to-run repeatability of the gain histories, the sensitivity of SEGA's sensor to channel-mean shifts at artificial states, attention-operator approximations and native-resolution semantic consistency. None bears on the claims of this paper, and we do not report them.

G Compute and timing

Table G.1 gives the sampling cost of the 1558 retained 4096^2 images, 79 GPU-hours in total (73 on GH200, 6 on GB200), with a median peak allocation of 40.8 GB. The second campaign's dev runs and the held-out SEGA + TM run used one GB200 node; every other run used GH200 nodes, and every forced-FlashAttention-2 timing at 4096^2 comes from the eight-prompt pilot. The GPU of each image was recovered from task and fleet logs, and its attention policy from the code revision pinned by its implementation hash, because the records carry a constant attention label (Section 5.5). The SDPA

Table G.1: Sampling cost is set by the attention kernel and the GPU, not by the method. Seconds per image cover prompt encoding, 28 denoising steps and VAE decoding; IQR is the interquartile range; peak is `torch.cuda.max_memory_allocated`; GPU-h sums the per-image seconds; *n* counts images. SDPA policies, method families and the separately listed node are defined in the text.

SDPA	Methods	*n*	Nodes	Median (s)	IQR (s)	Peak (GB)	GPU-h
4096^2, *GH200 (10 nodes)*							
FA2, forced	NTK	8	1	289.8	289.8–289.9	40.8	0.6
FA2, forced	SEGA family	23	3	289.8	289.7–290.1	40.8	1.9
FA2, forced	Constancy family	16	2	290.4	289.9–290.5	40.8	1.3
cuDNN first	NTK	160	6	187.2	186.2–187.5	40.8	8.3
cuDNN first	DyPE	167	6	187.9	187.6–188.1	40.8	8.7
cuDNN first	SEGA family	197	9	187.8	187.5–188.2	40.8	10.3
cuDNN first	SEGA + TM family	24	2	189.4	184.9–189.6	40.8	1.3
cuDNN first	Constancy family	532	8	187.9	187.6–188.6	40.8	27.8
4096^2, *GH200 node c619-102 (listed separately)*							
FA2, forced	YaRN, DyPE	16	1	329.6	329.5–329.7	40.8	1.5
cuDNN first	all methods	195	1	214.6	214.0–215.3	40.8	11.6
4096^2, *GB200 (1 node, 4 GPUs)*							
cuDNN first	SEGA + TM family	204	1	93.3	92.5–94.3	40.8	5.3
cuDNN first	Constancy family	16	1	93.2	92.1–93.3	40.8	0.4
native 1024^2, *GH200, FLUX.1-Krea-dev*							
FA2, forced	no RoPE change	16	1	5.47	5.47–5.48		
cuDNN first	no RoPE change	8	2	4.94	4.93–5.93	36.3	
4096^2, *GH200, official DyPE script*							
default	DyPE (official code)	1	1	193.3			

column of Table G.1 gives the requested backend policy (cuDNN first: cuDNN, then FlashAttention, then memory-efficient); the dispatched kernel was not logged. The SEGA family comprises released and paper-equation SEGA with their static ablations; the SEGA + TM family, SEGA + TM and its dev-only allocation controls; the Constancy family, Constancy and its dev sweeps, Constancy-TM, the temperature-only ablations, Alias-T and Alias-T-s. Node c619-102 ran 14% slower than the other GH200 nodes under both kernels (median power 665 W versus 836 W while generating), so it is listed separately; the forced-FA2 YaRN and DyPE pilots ran only there. The native rows run FLUX.1-Krea-dev without RoPE changes; the first image of each worker process carries a one-off warm-up (4.1 s with cuDNN first, 0.9 s with FA2), which widens the cuDNN-first IQR at 1024^2. The official DyPE script uses the default SDPA dispatch of diffusers. Timings exclude model loading (89 s per worker process) and PNG writing, except the single official DyPE run.

H Reproducibility checklist

- **Numbers.** `paper/tables/make_tables.py` recomputes every held-out paired difference from the archived per-prompt scores, checks it against the registered analysis output (largest absolute difference below 10^{-15}), and writes every table and every number cited in the text. `compute_table.py` and `mechanism_numbers.py` do the same for timings and mechanism statistics. Scratch inputs are snapshotted next to the scripts.
- **Protocol.** Dev rows 0–7 and held-out rows 8–194 of the Aesthetic-4K evaluation captions; seed 0; seed-1 replication on rows 8–62; selection gate and its outcomes in Appendix D.2.
- **Generation.** Every image has a JSON sidecar with the full configuration, code revision, implementation hash, executed noise levels, per-step gains, wall time and peak memory.
- **Statistics.** Paired percentile bootstrap, 2,000 resamples, generator seed 0, with nominal per-metric intervals. Stars in Tables B.1 and B.4 mark approximate centered-bootstrap tests adjusted by Holm within each table; the other descriptive tables retain nominal intervals.
- **Third-party code.** SEGA and DyPE are re-implemented from their released code at pinned revisions with numerical parity tests; DyPE's official code was also run end to end for the checkpoint

diagnosis.

- **Not done.** No human study; one model family; one resolution; one seed for most arms; FID and KID too noisy at these sample sizes to rank methods.